A Finite Computer-Assisted Verification of Robin's Inequality

via Colossally Abundant Profiles, with Exact Prime-Power Residual Dynamics

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Abstract

We give an unconditional finite, computer-assisted verification of Robin's inequality

$$\sigma(n) < e^{\gamma} n \log\log n$$

for every integer

$$5041 \le n \le 10^{71\,000\,000\,000\,000\,000\,000\,000} = 10^{7.1\times 10^{22}}.$$

The proof first certifies the exact logarithmic Robin margin for every colossally abundant exponent profile whose largest prime factor lies between 3 299 and

$$164\,967\,000\,000\,000\,000\,000\,000 = 1.64967 \times 10^{23}.$$

It combines an exhaustive directed-interval enumeration of 3 341 978 finite profiles with an analytic prime-power reduction, a smoothed explicit formula, rigorous finite-height verification of the Riemann hypothesis, and an absolute estimate for the unverified zero tail. Robin's convexity proposition between consecutive colossally abundant numbers then transfers the endpoint result to every integer in the stated range.

Separately from that finite certificate, we retain the signs discarded by the absolute high-zero estimate. We derive an exact recombination of the integrated prime residual with the deterministic buffer, isolate a continuous spectral residual $\mathfrak{B}(t)$, and prove its prime-power cell equation. The one-sided quantity $2+\gamma-\log(4\pi)-\mathfrak{B}(t)$ admits a first-crossing cone whose smooth flow replenishes a scalar reserve and whose prime-power events withdraw exactly $\Lambda(q)/\sqrt{q}$. Eventual nonnegativity of the one-sided quantity is RH-equivalent; the cone turns it into a precise RH-sufficient eventwise target rather than merely a request for a tighter absolute envelope. The positivity principle is of known Weil–Suzuki type; the reformulation developed here is its exact prime-side cell and kick dynamics. We further derive an exact signed triangular identity that reconstructs the post-event slope from an adaptive window of additive size $\asymp \sqrt{q}\log q$. It isolates a sharp combined prime-side inequality and a modular Turán-type route. The latter is RH-conditional and cannot be used as an unconditional proof step. A binary64 scan over all 5 762 859 prime-power events through 10^8 supports the finite pattern, but neither universal eventwise target is proved here.

All calculations used in the two finite theorems use deterministic sieves, exact integer arithmetic, and outward-rounded arbitrary-precision decimal intervals. The source and report hashes are recorded. The conclusion is finite: it neither proves Robin's inequality for all integers nor proves the Riemann hypothesis.

1 Introduction and precise scope

For $n \ge 1$, let

$$\sigma(n) = \sum_{d|n} d$$

and let γ be Euler's constant. Robin proved that the Riemann hypothesis is equivalent to

$$\sigma(n) < e^{\gamma} n \log\log n \qquad (n > 5040)$$

[1]. This paper does not resolve the infinite quantifier in Robin's criterion. Its first purpose is to isolate and prove a large explicit finite theorem; its second is to formulate the remaining signed problem as an exact prime-power dynamical target.

Define the exact logarithmic Robin margin

$$\mathcal{G}(n) := \gamma + \log\log\log n - \log \frac{\sigma(n)}{n}. \tag{1.1}$$

For $n > e^e$, Robin's inequality is equivalent to $\mathcal{G}(n) > 0$.

An integer C is *colossally abundant* (CA) if there exists $\varepsilon > 0$ such that

$$\frac{\sigma(m)}{m^{1+\varepsilon}} \le \frac{\sigma(C)}{C^{1+\varepsilon}} \qquad (m \ge 1).$$

At a transition value of ε, several integers can maximize the same objective. Every such tied maximizer is included in the term *CA profile* below. We use the classical multiplicative description and threshold structure of CA numbers due to Alaoglu and Erdős [2].

The two main results are as follows.

Theorem 1 (Finite CA theorem)**.** *Let C be any CA profile with exact prime support*

$$3\,299 \le P(C) \le 164\,967\,000\,000\,000\,000\,000\,000,$$

where $P(C)$ is the largest prime factor of C. Then

$$\mathcal{G}(C) > 0.$$

More precisely, with $x = P(C)$,

$$\sqrt{x}\log x\,\mathcal{G}(C) > 2 \times 10^{-7}.$$

Theorem 2 (Finite all-integer theorem)**.** *For every integer n satisfying*

$$5041 \le n \le 10^{71\,000\,000\,000\,000\,000\,000\,000} = 10^{7.1\times 10^{22}},$$

one has

$$\sigma(n) < e^{\gamma} n \log\log n.$$

Theorem 1 is proved by two overlapping mechanisms. A direct all-profile interval computation covers

$$3\,299 \le P(C) \le 56\,048\,351.$$

An analytic residual certificate, uniform over every CA exponent profile, covers

$$56\,048\,351 \le P(C) \le 164\,967\,000\,000\,000\,000\,000\,000.$$

Theorem 2 is then obtained from a direct base interval, an exact CA endpoint overlap, the complete intervening CA chain, Robin's interpolation proposition, and an explicit conversion from prime support to the size of a CA integer.

After proving these finite statements, Section 8 returns to the signed residual before absolute values are taken. It gives an exact prime-side dynamical formulation of the post-cutoff obstruction. None of the exploratory numerical observations in that section is used in Theorems 1 or 2.

This finite certificate is logically separate from the RH-strength claims audited in the corrected status manuscript [17]. In particular, the two earlier manuscripts are not altered or silently used as proofs of the finite numerical assertions made here.

The proofs of the two finite theorems use only the exact identity (3.6), which follows by direct algebra. They use neither the MVDC centre $K(Y, x)$, the reciprocal-prime route, nor any MVDC estimate.

Remark 3 (Meaning of computer-assisted)**.** No random sampling is used. The finite profile computation exhausts every support and every transition branch in its domain. The analytic continuation uses monotonicity on closed segments and directed interval endpoints. Nevertheless, the executable certificates are not proof-assistant kernels: their implementations and documented runtime semantics belong to the trusted base.

2 The exact Robin ledger on full prime support

Let

$$\beta(x) = \prod_{p \le x} \frac{p}{p-1}, \qquad E(x) = \log \beta(x) - \gamma - \log\log x.$$

For an integer with full support through x,

$$n = \prod_{p \le x} p^{a_p}, \qquad a_p \ge 1,$$

put

$$A(n,x) = -\sum_{p \le x} \log\left(1 - p^{-(a_p+1)}\right)$$

and

$$B_{\log}(n,x) = \log \frac{\log\log n}{\log x}.$$

The factorization

$$\frac{\sigma(n)}{n} = \beta(x) \prod_{p \le x} \left(1 - p^{-(a_p+1)}\right)$$

gives the following exact identity.

Lemma 4 (Exact Robin gate)**.** *For every full-support n as above,*

$$\mathcal{G}(n) = A(n,x) + B_{\log}(n,x) - E(x). \tag{2.1}$$

Consequently Robin's inequality is equivalent to the positivity of the right-hand side.

Proof. Taking logarithms in the product for $\sigma(n)/n$ gives

$$\log \frac{\sigma(n)}{n} = \gamma + \log\log x + E(x) - A(n,x).$$

On the other hand,

$$\gamma + \log\log\log n = \gamma + \log\log x + B_{\log}(n,x).$$

Subtracting proves (2.1). □

For later use, define

$$\theta(x) = \sum_{p \le x} \log p, \qquad \psi(x) = \sum_{p^m \le x} \log p,$$

$$H(n,x) = \log n - \theta(x), \qquad z(n,x) = \frac{\log n - x}{x}, \qquad L = \log x.$$

Then

$$\log\log n = L + \log(1+z),$$

so the nonlinear bridge remainder is

$$\mathcal{B}_2(n,x) = \log\left(1 + \frac{\log(1+z(n,x))}{L}\right) - \frac{z(n,x)}{L}. \tag{2.2}$$

3 Exact prime-power reduction

Put

$$F(t) = \frac{1}{t \log t}, \qquad \omega(t) = -F'(t) = \frac{\log t + 1}{t^2 \log^2 t},$$

and define the smoothed prime-distribution residual

$$\mathcal{R}_\infty(x) = \int_x^\infty (\psi(t) - t)\omega(t)\,\mathrm{d}t. \tag{3.1}$$

Its convergence is justified in Section 5.3, where the integrated zero series is shown to be absolutely convergent.

Let

$$\mathcal{H}_\Lambda(x) = \sum_{2 \le n \le x} \frac{\Lambda(n)}{n \log n} = \sum_{p^m \le x} \frac{1}{mp^m}.$$

Lemma 5 (Finite prime-side form of the residual)**.** *For every real $x > 1$, with either consistent one-sided convention at a prime-power endpoint,*

$$\boxed{\mathcal{R}_\infty(x) = \frac{\psi(x) - x}{x \log x} + \gamma + \log\log x - \mathcal{H}_\Lambda(x).} \tag{3.2}$$

The right-hand side is continuous at every prime power.

Proof. Stieltjes integration up to $Y > x$ gives

$$\begin{aligned}\int_x^Y (\psi(t) - t)\omega(t)\,\mathrm{d}t = {} & \frac{\psi(x) - x}{x \log x} - \frac{\psi(Y) - Y}{Y \log Y} \\ & + \sum_{x < n \le Y} \frac{\Lambda(n)}{n \log n} - \log\log Y + \log\log x.\end{aligned}$$

The PNT removes the boundary term, while $\mathcal{H}_\Lambda(Y) = \log\log Y + \gamma + o(1)$. Letting Y tend to infinity proves (3.2). At $x = p^m$, the endpoint term jumps by $1/(mp^m)$, exactly as does $\mathcal{H}_\Lambda$. □

Also define

$$R_{\mathrm{core}}(n, x) = \frac{H(n, x)}{x \log x} - \sum_{p \le x} \log\left(1 - p^{-(a_p+1)}\right). \tag{3.3}$$

3.1 The finite prime-power correction

Let

$$\mathcal{P}_\infty(x) = \int_x^\infty (\psi(t) - \theta(t))\omega(t)\,\mathrm{d}t$$

and

$$\mathcal{T}_2(x, \infty) = \sum_{p > x} \sum_{m \ge 2} \frac{1}{mp^m}.$$

Since $\psi - \theta \ge 0$, Tonelli's theorem and

$$\log p\, F(p^m) = \frac{1}{mp^m}$$

give

$$\mathcal{P}_\infty(x) = \frac{\psi(x) - \theta(x)}{x \log x} + \sum_{\substack{p, m \ge 2 \\ p^m > x}} \frac{1}{mp^m}.$$

The bases $p > x$ cancel on subtracting $\mathcal{T}_2$. Thus

$$\mathcal{C}_{\mathrm{pp}}(x) := \mathcal{P}_\infty(x) - \mathcal{T}_2(x,\infty) = \frac{\psi(x)-\theta(x)}{x\log x} + \sum_{\substack{p\le x,\ m\ge 2\\ p^m > x}} \frac{1}{mp^m}. \tag{3.4}$$

This is a finite prime-power quantity.

Let

$$W(x) = E(x) - \frac{\theta(x)-x}{x\log x}.$$

Partial summation of the first-order prime sum, followed by the convergent Euler-product tail, yields

$$W(x) = \int_x^\infty (t-\theta(t))\omega(t)\,\mathrm{d}t - \mathcal{T}_2(x,\infty).$$

Since

$$\mathcal{P}_\infty(x) - \mathcal{R}_\infty(x) = \int_x^\infty (t-\theta(t))\omega(t)\,\mathrm{d}t,$$

we obtain the exact relation

$$W(x) = \mathcal{C}_{\mathrm{pp}}(x) - \mathcal{R}_\infty(x). \tag{3.5}$$

Proposition 6 (Exact Robin-gap identity)**.** *For every full-support exponent vector,*

$$\boxed{\mathcal{G}(n) = \mathcal{R}_\infty(x) + \mathcal{B}_2(n,x) + R_{\mathrm{core}}(n,x) - \mathcal{C}_{\mathrm{pp}}(x).} \tag{3.6}$$

Proof. By (2.2),

$$B_{\log}(n,x) = \frac{z(n,x)}{L} + \mathcal{B}_2(n,x).$$

Moreover,

$$z(n,x) = \frac{\theta(x)-x+H(n,x)}{x}.$$

Substituting these identities and

$$E(x) = \frac{\theta(x)-x}{xL} + W(x)$$

into Lemma 4 cancels the endpoint term $(\theta(x)-x)/(xL)$. Equation (3.5) then gives (3.6). □

3.2 A positive profile-independent core

For $p \le x$, let

$$k_p(x) = \max\{k \ge 1 : p^k \le x\},$$

$$c_{p,x} = \frac{\log p}{x\log x}, \qquad d_{p,a} = -\log\left(1 - p^{-(a+1)}\right),$$

and

$$\Phi_{p,x}(a) = (a-1)c_{p,x} + d_{p,a}.$$

Then

$$R_{\mathrm{core}}(n,x) = \sum_{p\le x} \Phi_{p,x}(a_p).$$

Also put

$$Q_{p,k} = \sum_{m=k+1}^{\infty} \frac{1}{mp^m}.$$

Equation (3.4) becomes

$$\mathcal{C}_{\mathrm{pp}}(x) = \sum_{p\le x} \left((k_p - 1)c_{p,x} + Q_{p,k_p}\right). \tag{3.7}$$

Lemma 7 (Strict prime-by-prime domination)**.** *For every real $x \geq 2$, every prime $p \leq x$, and every integer $a \geq 1$,*

$$\Phi_{p,x}(a) > (k_p - 1)c_{p,x} + Q_{p,k_p}.$$

Proof. Write

$$\delta_{p,a} = d_{p,a} - d_{p,a+1} = \log\left(1 + \frac{1}{p + p^2 + \cdots + p^{a+1}}\right).$$

The sequence $\delta_{p,a}$ is strictly decreasing and

$$\Phi_{p,x}(a+1) - \Phi_{p,x}(a) = c_{p,x} - \delta_{p,a}.$$

Let $k = k_p$. If $a < k$, then $k \geq 2$, and

$$\delta_{p,k-1} > \frac{1}{1 + p + \cdots + p^k} \geq \frac{1}{kp^k} \geq c_{p,x}.$$

Here $1+p+\cdots+p^k < 2p^k \leq kp^k$, while $x \log x \geq kp^k \log p$. Thus $\Phi_{p,x}$ decreases strictly through index k, and $\Phi_{p,x}(a) > \Phi_{p,x}(k)$.

For $a \geq k$, set

$$u = \frac{1}{(k+1)p^{k+1}}.$$

Since $p^k \leq x < p^{k+1}$,

$$c_{p,x} > u, \qquad Q_{p,k} \leq \frac{u}{1 - 1/p}.$$

If $a = k$, then

$$d_{p,k} > p^{-(k+1)} = (k+1)u \geq \frac{u}{1 - 1/p}.$$

If $a = k + 1$, then

$$c_{p,x} + d_{p,k+1} > u + \frac{1}{p^{k+2}} = u\left(1 + \frac{k+1}{p}\right) \geq \frac{u}{1 - 1/p}.$$

If $a \geq k + 2$, then

$$(a - k)c_{p,x} > 2u \geq \frac{u}{1 - 1/p}.$$

These cases prove the claim. □

Define the optimal full-support core and its finite buffer by

$$R_*(x) = \sum_{p \leq x} \min_{a \geq 1} \Phi_{p,x}(a), \qquad D_*(x) = R_*(x) - \mathcal{C}_{\mathrm{pp}}(x). \tag{3.8}$$

Corollary 8. *For every $x \geq 2$,*

$$R_{\mathrm{core}}(n, x) - \mathcal{C}_{\mathrm{pp}}(x) \geq D_*(x) > 0.$$

Consequently it is sufficient to prove

$$\mathcal{R}_\infty(x) + \mathcal{B}_2(n, x) > -D_*(x). \tag{3.9}$$

The minimizing exponent is characterized by

$$a_p^*(x) = \min\left\{a \geq 1 : \delta_{p,a} \leq \frac{\log p}{x \log x}\right\}.$$

If equality occurs, the adjacent exponents have the same value of $\Phi_{p,x}$; hence D_* is tie-safe. Notice that this full-support optimizer is not the CA-family activation rule $\tau_s(p) > \tau_1(x)$.

4 Certified lower bounds for the deterministic buffer

4.1 Complete finite sweep

Proposition 9 (Finite D_* certificate)**.** *At every one of the* 3 340 551 *prime supports*

$$3\,299 \le x \le 56\,048\,351,$$

the directed interval certificate proves

$$\sqrt{x}\log x\, D_*(x) > 0.7825.$$

The certified global minimum lower endpoint is

$$0.782599918837756405719105563315190371695 7$$

at $x = 56\,048\,351$.

The verifier uses a deterministic odd-only sieve, exact integer prime-power events, outward-rounded logarithms and square roots, and rigorous alternating series bounds above a fixed exact-log cutoff. It reports no failed or inconclusive support. Direct reconstructions at $3299, 5297, 5303, 10007$ overlap the streamed intervals; the pair $5297, 5303$ straddles a change between the CA-only and full-support activation conventions.

4.2 Analytic continuation beyond the sieve

Put

$$y = \sqrt{2x}, \qquad L = \log x, \qquad \ell = \log y = \frac{L + \log 2}{2}.$$

For $y < p \le x$, one has $p^2 > x$, and the exact optimizer is $a_p^*(x) = 1$. Indeed,

$$\frac{\delta_{p,1}}{\log p} = \frac{1}{\log p} \log\left(1 + \frac{1}{p + p^2}\right) < \frac{1}{p^2 \log p} < \frac{1}{x \log x},$$

where the last inequality follows from $p^2 > 2x$ and $\log p > \frac{1}{2}\log x$. The characterization preceding Proposition 9 now gives the claim. The corresponding contribution to $D_*(x)$ is

$$d_{p,1} - Q_{p,1} = \frac{1}{p} - \log\left(1 + \frac{1}{p}\right) > \left(\frac{1}{2} - \frac{1}{3y}\right)\frac{1}{p^2}.$$

All omitted prime contributions are positive.

Use Dusart's explicit estimate

$$|\theta(t) - t| < \frac{3.965t}{\log^2 t} \qquad (t \ge 2) \tag{4.1}$$

[4]. Partial summation over $y < p \le x$ gives the following closed lower expression. Put

$$\epsilon = \frac{3.965}{\ell^2}, \qquad q(x) = 1 - \sqrt{\frac{2}{x}},$$

$$A(L) = \frac{L}{\sqrt{2}\,\ell}, \qquad \underline{J}(L) = \frac{L}{\sqrt{2}}\left(\frac{q(x)}{\ell} - \frac{1}{\ell^2}\right),$$

$$\underline{S}(L) = (1-\epsilon)\underline{J}(L) - 2\epsilon A(L) - \frac{1}{\sqrt{x}},$$

and

$$D_{\mathrm{lb}}(x) = \left(\frac{1}{2} - \frac{1}{3\sqrt{2x}}\right)\underline{S}(L).$$

Proposition 10 (Analytic D_* continuation)**.** *For every real $x \geq 56\,048\,351$,*

$$\sqrt{x}\log x\, D_*(x) > D_{\mathrm{lb}}(x) > 0.5.$$

Moreover D_{lb} is increasing,

$$D_{\mathrm{lb}}(56\,048\,351) > 0.516114369552624825734994861539,$$

and

$$D_{\mathrm{lb}}(164\,967\,000\,000\,000\,000\,000\,000) > 0.661088880265812593800990356530.$$

Proof. Put

$$S_2(y,x) = \sum_{y<p\leq x} \frac{1}{p^2}.$$

Partial summation gives

$$\begin{aligned} S_2(y,x) = {} & \frac{\theta(x)}{x^2 L} - \frac{\theta(y)}{y^2 \ell} \\ & + \int_y^x \theta(t)\left(\frac{2}{t^3\log t} + \frac{1}{t^3\log^2 t}\right)\,\mathrm{d}t. \end{aligned}$$

For $t \in [y,x]$, equation (4.1) implies $(1-\epsilon)t < \theta(t) < (1+\epsilon)t$. Dropping the nonnegative endpoint at x, using the upper estimate at the negative y-endpoint, and using the lower estimate in the integral yields

$$S_2(y,x) > -\frac{1+\epsilon}{y\ell} + (1-\epsilon)\left(\frac{1}{y\ell} - \frac{1}{xL} + J(y,x)\right),$$

where

$$J(y,x) = \int_y^x \frac{\mathrm{d}t}{t^2\log t}.$$

Writing $r = x/y = \sqrt{x/2}$, substituting $t = yu$, and using

$$\frac{1}{\ell + \log u} \geq \frac{1}{\ell} - \frac{\log u}{\ell^2} \qquad (u \geq 1)$$

gives

$$J(y,x) > \frac{1}{y}\left(\frac{1-1/r}{\ell} - \frac{1}{\ell^2}\right).$$

Since $1 - 1/r = q(x)$, multiplication by $\sqrt{x}L$ gives

$$\sqrt{x}LS_2(y,x) > \underline{S}(L).$$

Here replacing $-(1-\epsilon)/\sqrt{x}$ by $-1/\sqrt{x}$ only weakens the lower bound. The prefactor $1/2 - 1/(3\sqrt{2x})$ comes from the alternating logarithm series. For monotonicity, ϵ decreases; q, L/ℓ, and $\underline{J}$ increase; $\epsilon A = 3.965L/(\sqrt{2}\,\ell^3)$ decreases because its derivative has the sign of $\log 2 - 2L$; and both $-x^{-1/2}$ and $1/2 - 1/(3\sqrt{2x})$ increase. The factors $1-\epsilon$, $\underline{J}$, and the resulting $\underline{S}$ are positive at $56\,048\,351$. Hence $\underline{S}$ and D_{lb} are positive and increasing. The endpoint evaluations are directed intervals. The 60- and 80-digit enclosures are nested. $\square$

5 Uniform control of the signed residual

5.1 Tie-safe CA layer bounds

For $s \geq 1$, define the CA transition threshold

$$\tau_s(p) = \frac{1}{\log p} \log \frac{1 - p^{-(s+1)}}{1 - p^{-s}}.$$

If x is the exact support and q is the next prime, the CA parameter satisfies

$$\tau_1(q) \leq \varepsilon \leq \tau_1(x).$$

If $a_p \geq s \geq 2$, then $\varepsilon \leq \tau_s(p)$. Since

$$\tau_1(q) > \frac{1}{(q+1)\log q}, \qquad \tau_s(p) < \frac{1}{p^s \log p},$$

one has $p^s \log p < (q+1)\log q$. If $p < q^{1/s}$, the desired bound is immediate. Otherwise $\log q / \log p \leq s$, and the same inequality again gives $p^s < s(q+1)$. Thus one obtains the strict, tie-safe implication

$$a_p \geq s \quad \Longrightarrow \quad p < (s(q+1))^{1/s}. \tag{5.1}$$

Hence

$$H(n,x) = \sum_{s \geq 2} \sum_{a_p \geq s} \log p \leq \sum_{\substack{s \geq 2 \\ (s(q+1))^{1/s} > 2}} \theta\left((s(q+1))^{1/s}\right).$$

Dusart's short-prime theorem [3] gives, for $x \geq 3275$,

$$q \leq x \left(1 + \frac{1}{2\log^2 x}\right).$$

Together with Rosser–Schoenfeld's global estimate $\theta(t) < 1.01624t$ [5], the decreasing layer majorant gives

$$H(n,x) < 2.035\sqrt{x} \qquad (x \geq 56\,048\,351) \tag{5.2}$$

through the range used here. On the lower interval an event-by-event directed analysis gives

$$H(n,x) < 2.634\sqrt{x} \qquad (3\,299 \leq x \leq 56\,048\,351). \tag{5.3}$$

The latter checks the left endpoint and the right limits of the fifteen layer activation events $s = 16, \ldots, 30$; between events the normalized majorant decreases.

The largest layer cutoff at $x = 164\,967\,000\,000\,000\,000\,000\,000$ is

$$< 5.7445 \times 10^{11} < 10^{19}.$$

Thus Büthe's finite theorem [7] can alternatively be used at every layer cutoff, where it supplies $\theta(u) < u$.

5.2 The nonlinear bridge

Büthe's partial-RH estimate [6] states that, for $x \geq 5000$,

$$|\theta(x) - x| \leq \frac{\sqrt{x}\log x(\log x - 2)}{8\pi}, \tag{5.4}$$

provided RH has been verified through height T and

$$4.92\sqrt{\frac{x}{\log x}} \leq T.$$

At $x = 164\,967\,000\,000\,000\,000\,000\,000$, the required height is less than

$$273\,305\,915\,796 < 3 \times 10^{12}.$$

Platt and Trudgian rigorously verified RH through 3×10^{12} [8]; hence (5.4) holds throughout the upper range.

Combining (5.2) with (5.4) gives

$$|z(n,x)| < \delta(L) := \left(\frac{L(L-2)}{8\pi} + 2.035\right) e^{-L/2}.$$

The directed endpoint and derivative checks give $0 < \delta(L) < 0.001774 < 1$ throughout the upper range. For

$$b_L(z) = \log\left(1 + \frac{\log(1+z)}{L}\right) - \frac{z}{L},$$

one has $b_L(0) = b_L'(0) = 0$ and

$$b_L''(z) = -\frac{A+1}{(1+z)^2 A^2}, \qquad A = L + \log(1+z).$$

Taylor's theorem at $z = 0$, evaluated at the worst lower endpoint $-\delta(L)$, gives a directed lower bound for $\mathcal{B}_2$. Its normalized upper loss is the explicit function

$$C_{\mathcal{B}_2}(x) = \frac{\sqrt{x}L\,\delta(L)^2}{2} \frac{A_-(L)+1}{(1-\delta(L))^2 A_-(L)^2}, \qquad A_-(L) = L + \log(1-\delta(L)), \tag{5.5}$$

so that

$$-\sqrt{x}L\,\mathcal{B}_2(n,x) \le C_{\mathcal{B}_2}(x).$$

This function decreases over the upper range. At $x = 164\,967\,000\,000\,000\,000\,000\,000$,

$$-\sqrt{x}L\,\mathcal{B}_2(n,x) < 1.5589734604839492 \times 10^{-8}. \tag{5.6}$$

5.3 Integrated explicit formula

Let $\rho = \beta + i\gamma_\rho$ run over nontrivial zeros of $\zeta(s)$, with multiplicity. Starting from the standard symmetric truncation of the von Mangoldt explicit formula [10, Ch. 17], insert a compact upper cutoff and integrate the finite zero sum against ω. After the substitution $t = e^u$, put

$$h(u) = \frac{u+1}{u^2}, \qquad J_\rho(L) = \int_L^\infty e^{-(1-\rho)u} h(u)\,\mathrm{d}u.$$

For $\Re z > 0$, write

$$E_1(z) = \int_1^\infty \frac{e^{-zu}}{u}\,\mathrm{d}u.$$

One direct integration gives the exact closed form

$$\boxed{J_\rho(L) = \frac{e^{-(1-\rho)L}}{L} + \rho E_1((1-\rho)L).} \tag{5.7}$$

The half-weight convention for ψ at a jump is immaterial here, because changing values at the prime-power endpoints does not change the integral. Passing first through symmetric zero truncations and then through compact upper cutoffs gives

$$\mathcal{R}_\infty(x) = -\sum_\rho \frac{J_\rho(L)}{\rho} - \frac{\log(2\pi)}{xL} - \frac{1}{2}\int_x^\infty \log(1-t^{-2})\omega(t)\,\mathrm{d}t. \tag{5.8}$$

The last term is positive and may be discarded in a lower bound.

The limiting passage is not justified merely by writing a formal termwise integral. Two integrations by parts below give

$$\left|\frac{J_\rho(L)}{\rho}\right| = O_L(\gamma_\rho^{-2}).$$

The zero count $N(T) = O(T\log T)$ then makes the integrated zero series absolutely convergent, completing the cutoff argument.

The exact smoothing performed by the integral is visible after multiplying the zero part by its natural Robin scale. Define

$$A_\rho(L) = \frac{1}{\rho} + Le^{(1-\rho)L}E_1((1-\rho)L).$$

Then

$$e^{L/2}L\left(-\sum_\rho \frac{J_\rho(L)}{\rho}\right) = -\sum_\rho e^{(\rho-1/2)L}A_\rho(L). \tag{5.9}$$

The identity

$$e^z E_1(z) = \int_0^\infty \frac{e^{-v}}{z+v}\,\mathrm{d}v$$

and a two-term geometric expansion give the rigorous remainder formula

$$\boxed{A_\rho(L) = \frac{1}{\rho(1-\rho)} - \frac{1}{(1-\rho)^2L} + R_{\rho,2}(L), \qquad |R_{\rho,2}(L)| \le \frac{2}{|1-\rho|^3L^2}.} \tag{5.10}$$

Thus the endpoint Chebyshev error and the logarithmically weighted Mangoldt error cancel their separate $1/\gamma_\rho$ coefficients and leave an absolutely summable $1/\gamma_\rho^2$ kernel. They do not cancel the factor $e^{(\beta-1/2)L}$. A hypothetical zero with $\beta > 1/2$ therefore remains an exponentially growing normalized mode. This distinction is the reason that sharper absolute estimates can extend the finite certificate but cannot by themselves settle the infinite problem.

5.4 Verified low zeros

Let $T = 3 \times 10^{12}$. By the Platt–Trudgian theorem cited above, every zero with $|\gamma_\rho| \le T$ has $\rho = 1/2 + i\gamma_\rho$. For such a zero, one integration by parts yields

$$|J_\rho(L)| \le \frac{x^{-1/2}}{|1-\rho|}\left(h(L) + \frac{2(L+2)}{L^3}\right).$$

The exact identity

$$\sum_\rho \frac{1}{\rho(1-\rho)} = 2 + \gamma - \log(4\pi)$$

bounds the verified subset. Omitted zero pairs have positive real contribution to this identity because

$$\mathrm{Re}\big(\rho(1-\rho)\big) = \beta(1-\beta) + \gamma_\rho^2 > 0.$$

Consequently the normalized low-zero loss, including the $\log(2\pi)$ term, is at most

$$C_{\mathrm{low}}(x) = (2 + \gamma - \log(4\pi))\left(1 + \frac{3}{L} + \frac{4}{L^2}\right) + \frac{\log(2\pi)}{\sqrt{x}}. \tag{5.11}$$

This decreases with x.

5.5 Unverified high zeros

Twice integrating J_ρ by parts gives

$$J_\rho(L) = e^{-(1-\rho)L}\left(\frac{h(L)}{1-\rho} + \frac{h'(L)}{(1-\rho)^2}\right) + \frac{1}{(1-\rho)^2}\int_L^\infty e^{-(1-\rho)u} h''(u)\,\mathrm{d}u.$$

Here

$$h'(u) = -\frac{1}{u^2} - \frac{2}{u^3}, \qquad h''(u) = \frac{2}{u^3} + \frac{6}{u^4} > 0.$$

It follows that

$$\left|\frac{J_\rho(L)}{\rho}\right| \le x^{\beta-1}\left(\frac{h(L)}{\gamma_\rho^2} + \frac{2[-h'(L)]}{|\gamma_\rho|^3}\right). \tag{5.12}$$

For every positive ordinate, the functional equation gives the multiplicity-preserving involution

$$\beta + i\gamma_\rho \longmapsto 1 - \beta + i\gamma_\rho.$$

Convexity gives

$$x^{\beta-1} + x^{-\beta} \le 1 + x^{-1}.$$

Consequently, for every nonnegative ordinate weight w_γ, averaging over this involution yields

$$\sum_{\rho:\ \gamma_\rho > T} x^{\beta-1} w_{\gamma_\rho} \le \frac{1+x^{-1}}{2} \sum_{\rho:\ \gamma_\rho > T} w_{\gamma_\rho}.$$

The fixed points $\beta = 1/2$ satisfy the same inequality by $x^{-1/2} \le (1+x^{-1})/2$. Conjugation then doubles both sides and includes the negative ordinates. Thus the accounting is valid for on-line zeros, off-line quartets, and all multiplicities.

Retain $T = 3 \times 10^{12}$, and define positive-ordinate tails

$$S_k(T) = \sum_{\rho:\ \gamma_\rho > T} \gamma_\rho^{-k},$$

where zeros are counted with multiplicity. Writing $N(t)$ for the number of zeros with $0 < \gamma_\rho \le t$, counted with multiplicity, Hasanalizade, Shen and Wong prove, for $t \ge e$,

$$\left|N(t) - \frac{t}{2\pi}\log\frac{t}{2\pi e}\right| \le a\log t + b\log\log t + c,$$

where

$$(a, b, c) = (0.1038, 0.2573, 9.3675)$$

[9]. Direct Stieltjes integration yields

$$\begin{aligned} S_2(T) \le{}& \frac{\log(T/(2\pi)) + 1}{2\pi T} \\ &+ \frac{2a\log T + a/2 + 2b\log\log T + b/(2\log T) + 2c}{T^2}. \end{aligned} \tag{5.13}$$

Directed evaluation gives

$$S_2(T) < 1.4797036309153468749461176940 \times 10^{-12}$$

and

$$S_3(T) \le \frac{S_2(T)}{T} < 4.9323454363844895831537256466 \times 10^{-25}.$$

Combining these estimates with (5.12) gives

$$C_{\mathrm{high}}(x) \le \sqrt{x}L(1+x^{-1})\left[h(L)S_2(T) + 2[-h'(L)]S_3(T)\right]. \tag{5.14}$$

The right-hand side is increasing over the certified range.

5.6 Segmented residual comparison

The analytic certifier uses 662 closed segments. On a segment $[a, b]$, monotonicity permits the lower comparison

$$D_{\mathrm{lb}}(a) - C_{\mathrm{low}}(a) - C_{B_2}(a) - C_{\mathrm{high}}(b). \tag{5.15}$$

Every directed segment interval has positive lower endpoint. The minimum, on the final segment

$$[1.64950 \times 10^{23}, 1.64967 \times 10^{23}],$$

is

$$2.1540734262173041165880364748 \times 10^{-7}. \tag{5.16}$$

At $x = 164\,967\,000\,000\,000\,000\,000\,000$, the same-point budget is

normalized component	directed bound
$D_{\mathrm{lb}}(x)$	$> 0.6610888802658125938009903565$
$C_{\mathrm{low}}(x)$	$< 0.0488481765327333219862574136$
$C_{B_2}(x)$	$< 1.5589734604839492 \times 10^{-8}$
$C_{\mathrm{high}}(x)$	$< 0.6122403598039197333285731567$
same-point clearance	$> 3.2833942493364667 \times 10^{-7}$

This proves (3.9) on the upper range.

6 Complete direct CA enumeration

For a fixed CA parameter ε, the objective is multiplicative and the prime exponents separate. Increasing the exponent of p from $s - 1$ to s changes the logarithm of the objective by

$$\log \frac{1 - p^{-(s+1)}}{1 - p^{-s}} - \varepsilon \log p.$$

Thus all changes occur at the exact thresholds $\tau_s(p)$. If q is the next prime after an exact support x, its first-layer cell is

$$\tau_1(q) \leq \varepsilon \leq \tau_1(x).$$

Every higher-layer threshold inside that cell creates two tied CA profiles, one on each side of the transition.

The all-profile verifier performs the following exhaustive construction:

1. generate the complete prime stream through $56\,048\,351$;
2. generate every relevant higher-layer event and prove that events beyond the search cutoff cannot enter the parameter range;
3. place each event into its unique first-layer support cell by directed threshold comparisons;
4. evaluate the exact logarithmic gap (1.1) before and after every event;
5. verify that no threshold relation is unresolved and that no simultaneous higher-event group was omitted.

Proposition 11 (Direct all-profile certificate)**.** *For every CA exponent profile with exact support*

$$3\,299 \leq x \leq 56\,048\,351,$$

the directed interval evaluation gives $\mathcal{G}(n) > 0$. The exhaustive stream contains

$$3\,340\,551 \text{ prime supports},$$

$$3\,341\,978 \text{ distinct numerical profiles},$$

and 1 427 *active higher-layer ties, with both branches checked. There are no failed, unresolved, or simultaneous higher-tie groups.*

The global streamed lower bound occurs at $x = 56\,048\,351$:

$$\mathcal{G}(n) > 5.836827418662993775771303247787 6 \times 10^{-6}.$$

A separate reconstruction at the same profile gives

$$\mathcal{G}(n) \in [5.8368302715897745 \times 10^{-6}, 5.8368766311571295 \times 10^{-6}],$$

which overlaps the streamed interval. The corresponding directly reconstructed normalized lower endpoint exceeds 0.7796421.

Proof of Theorem 1. Proposition 11 covers $3\,299 \leq x \leq 56\,048\,351$. On $56\,048\,351 \leq x \leq 164\,967\,000\,000\,000\,000\,000\,000$, Proposition 6, Corollary 8, and the segmented comparison (5.15) give

$$\sqrt{x}\log x\,\mathcal{G}(n) > 2.1540734262173041 \times 10^{-7}.$$

Both arguments include every exponent choice at a CA transition. Their domains overlap at $x = 56\,048\,351$, proving the theorem. ☐

7 From CA profiles to every integer

7.1 Robin's consecutive-CA interpolation

Proposition 12 (Robin)**.** *Let $C < C'$ be consecutive CA numbers and let*

$$f(n) = \frac{\sigma(n)}{n\log\log n}.$$

For $C < n < C'$,

$$f(n) \leq \max\{f(C), f(C')\}.$$

This is Proposition 1 in Section 3 of Robin's paper [1, p. 192]. For completeness, let ε be their common transition parameter. CA optimality gives

$$\frac{\sigma(n)}{n^{1+\varepsilon}} \leq \frac{\sigma(C)}{C^{1+\varepsilon}} = \frac{\sigma(C')}{(C')^{1+\varepsilon}}.$$

With $t = \log n$, the logarithm of the resulting upper envelope for $f(n)$ is a constant plus

$$\phi(t) = \varepsilon t - \log\log t.$$

Since

$$\phi''(t) = \frac{\log t + 1}{t^2\log^2 t} > 0,$$

the maximum on the closed interval is attained at an endpoint. If the two endpoint Robin inequalities are strict, the conclusion for all interior integers is strict.

7.2 The lower endpoint and the complete CA chain

The following three executable facts close the lower handoff.

1. Every one of the 50 400 integers

$$5041 \leq n \leq 55\,440$$

passes a directed interval check. The divisor sum is computed exactly both by a sieve and independently by smallest-prime-factor factorization. The minimum logarithmic gap occurs at $n = 10080$ and is greater than

$$0.01428290481768934155665716773379 79.$$

2. At the support-11 boundary, the old tied state is

$$2^4 3^2 5 \cdot 7 = 5040,$$

whereas exact support 11 forces

$$2^4 3^2 5 \cdot 7 \cdot 11 = 55\,440.$$

Directed comparisons against $\tau_1(11)$ exclude every other relevant layer ambiguity.

3. The complete small-support bridge $11 \leq x \leq 3299$ contains 459 prime supports, 503 numerical CA profiles, and 44 active higher-layer ties. Both branches of every tie are positive, with no failure or unresolved comparison.

These facts, Proposition 11, Theorem 1, and Proposition 12 certify every relevant CA endpoint with exact support from 11 through 164 967 000 000 000 000 000 000, together with every integer between consecutive covered endpoints. We now select a unique covered endpoint large enough that every preceding endpoint above the base overlap also has certified support.

7.3 Converting support into an integer cutoff

Let

$$X = 164\,967\,000\,000\,000\,000\,000\,000$$

and put

$$y = \left\lfloor \frac{999X}{1000} \right\rfloor = 164\,802\,033\,000\,000\,000\,000\,000.$$

Dusart's short-prime interval gives a prime q satisfying

$$y < q \leq y\left(1 + \frac{1}{2\log^2 y}\right).$$

Directed arithmetic proves

$$\frac{1}{2\log^2 y} < 0.000174955590614650 < \frac{1}{999},$$

and hence $q < X$.

Let q^+ be the successor of q. Choose

$$\tau_1(q^+) < \varepsilon_0 < \tau_1(q)$$

outside every higher-layer threshold in this cell. Such a choice exists: there are finitely many primes at most q, and for each one $\tau_s(p) \to 0$ as $s \to \infty$. The multiplicative threshold rule then gives a unique CA profile C_q of exact support q.

We also need the order of the entire preceding CA chain. Put

$$F(n) = \log \frac{\sigma(n)}{n}.$$

If $\varepsilon_1 > \varepsilon_2$ and C_1, C_2 are respective CA maximizers, their two optimality inequalities are

$$F(C_1) - \varepsilon_1 \log C_1 \geq F(C_2) - \varepsilon_1 \log C_2,$$

$$F(C_2) - \varepsilon_2 \log C_2 \geq F(C_1) - \varepsilon_2 \log C_1.$$

Adding them shows

$$(\varepsilon_1 - \varepsilon_2)(\log C_2 - \log C_1) \geq 0.$$

Thus CA size is nondecreasing as the parameter decreases. Since C_q is unique at ε_0, every CA number smaller than C_q belongs to a parameter greater than ε_0, and therefore has prime support at most $q < X$.

Finally, C_q has full prime support, so

$$\log C_q \geq \theta(q) > \theta(y).$$

Using the lower side of (4.1),

$$\theta(y) > y\left(1 - \frac{3.965}{\log^2 y}\right).$$

The directed evaluation is

$$\frac{y\left(1 - 3.965/\log^2 y\right)}{\log 10} > 71\,473\,313\,849\,371\,279\,821\,308.0122.$$

In particular,

$$C_q > 10^{71\,000\,000\,000\,000\,000\,000\,000}.$$

The natural-log margin after rounding the exponent down to $71\,000\,000\,000\,000\,000\,000\,000$ is greater than

$$1.0898454138699380786 57 \times 10^{21}.$$

Proof of Theorem 2. The direct base covers $5041 \leq n \leq 55\,440$, and the support-11 certificate identifies $55\,440$ as the identical endpoint of the CA bridge. The bridge, Theorem 1, and Robin's interpolation cover every integer through C_q, because the monotonicity argument above places every preceding CA endpoint inside the certified support range. The support-to-size estimate gives $C_q > 10^{71\,000\,000\,000\,000\,000\,000\,000}$, so every integer in the asserted interval is covered. ☐

8 Beyond the finite absolute envelope: exact signed residual dynamics

The finite theorems have now been proved. This section addresses a different question: what exact structure remains if the signs in the residual are kept instead of replacing the unverified zero tail by its absolute value? All identities and analytic reductions below are unconditional unless a statement explicitly invokes RH. The floating-point diagnostics at the end of the section are not part of the finite certificate.

8.1 Exact signed recombination with the deterministic buffer

Let

$$n_*(t) = \prod_{p\le t} p^{a_p^*(t)}, \qquad \Psi_*(t) = \log n_*(t),$$

where $a_p^*(t)$ is the minimizing exponent from (3.8). At an activation tie choose the new exponent, making Ψ_* right-continuous. This convention does not change any integral. Put

$$j_p = \frac{1}{p} - \log\left(1 + \frac{1}{p}\right) > 0.$$

Proposition 13 (Signed buffer identity)**.** *For every $x \ge 2$,*

$$\boxed{D_*(x) = \int_x^\infty (\Psi_*(t) - \psi(t))\omega(t)\,\mathrm{d}t - \sum_{p>x} j_p,} \tag{8.1}$$

and therefore

$$\boxed{\mathcal{R}_\infty(x) + D_*(x) = \int_x^\infty (\Psi_*(t) - t)\omega(t)\,\mathrm{d}t - \sum_{p>x} j_p.} \tag{8.2}$$

Equivalently,

$$\mathcal{R}_\infty(x) + D_*(x) = \gamma + \log\log x - \frac{1}{\log x} + \frac{\log n_*(x)}{x\log x} - \log\frac{\sigma(n_*(x))}{n_*(x)}. \tag{8.3}$$

Proof. For $s \ge 2$, let $\alpha_{p,s}$ be the point at which the minimizing exponent changes from $s-1$ to s. With F as above and

$$\delta_{p,s-1} = -\log(1 - p^{-s}) + \log(1 - p^{-(s+1)}),$$

the activation equation is

$$F(\alpha_{p,s}) = \frac{\delta_{p,s-1}}{\log p}, \qquad \frac{p^s}{s} < \alpha_{p,s} < p^s.$$

Prime-by-prime Stieltjes integration of all future optimizer activations telescopes to

$$\int_x^\infty (\Psi_*(t) - \theta(t))\omega(t)\,\mathrm{d}t = R_*(x) + \sum_{p>x} -\log(1 - p^{-2}).$$

The corresponding identity for the natural higher prime powers is

$$\int_x^\infty (\psi(t) - \theta(t))\omega(t)\,\mathrm{d}t = \mathcal{C}_{\mathrm{pp}}(x) + \sum_{p>x}\left[-\log(1 - p^{-1}) - \frac{1}{p}\right].$$

Subtracting gives (8.1), because the difference of the two displayed prime tails is j_p. Adding (3.1) proves (8.2). The finite form (8.3) follows by inserting the minimizing exponents in the separated core and using (3.2). $\square$

The correction satisfies

$$\sum_{p>x} j_p \sim \frac{1}{2x\log x},$$

so it is smaller than the Robin scale. Formula (8.2) is stronger structurally than separate lower bounds for $\mathcal{R}_\infty$ and D_*: the occurrence of ψ cancels exactly. It does not, by itself, supply a sign for the remaining weighted tail.

There is also a rigorous limitation on what the higher optimizer layers can cancel. For $\Re w > 1$, define the Mellin–Stieltjes transform of the jumps of Ψ_* by

$$\mathcal{M}_*(w) := \int_{1^-}^{\infty} u^{-w}\, \mathrm{d}\Psi_*(u) = -\frac{\zeta'(w)}{\zeta(w)} + H_*(w),$$

where

$$H_*(w) = \sum_p \sum_{s \geq 2} \log p \left(\alpha_{p,s}^{-w} - p^{-sw} \right).$$

The bounds on $\alpha_{p,s}$ imply local uniform convergence of H_* in $\Re w > 1/2$. Hence H_* is analytic there. Since $j_p = O(p^{-2})$, the Dirichlet series $\sum_p j_p p^{-w}$ is analytic for $\Re w > -1$. Consequently a pole of $-\zeta'/\zeta$ arising from a hypothetical zero with $\Re \rho > 1/2$ cannot be canceled by these higher layers. A universal argument needs a genuine signed first-layer mechanism.

The nonlinear bridge $\mathcal{B}_2$ is also too small to remove such a leading mode. Indeed, for

$$b_L(z) = \log \left(1 + \frac{\log(1+z)}{L} \right) - \frac{z}{L},$$

one has

$$b_L(0) = b_L'(0) = 0, \qquad b_L(z) = -\frac{L+1}{2L^2} z^2 + O(z^3).$$

If a first-layer mode has $z \asymp x^{\beta-1}$ with $\beta < 1$, then

$$\mathcal{B}_2 = O\left(\frac{x^{2\beta-2}}{\log x} \right) = o\left(\frac{x^{\beta-1}}{\log x} \right).$$

Thus neither the higher optimizer layers nor this quadratic bridge can erase the leading off-line mode.

8.2 The continuous spectral residual

Equation (5.10) isolates the leading long-scale zero coefficient. Define

$$\boxed{\mathfrak{B}(t) = \sum_\rho \frac{e^{(\rho-1/2)t}}{\rho(1-\rho)}, \qquad t \in \mathbb{R}.} \tag{8.4}$$

Because $N(T) = O(T \log T)$, the series converges absolutely and locally uniformly. Thus $\mathfrak{B}$ is an ordinary continuous function. Conjugation and the functional equation give

$$\mathfrak{B}(-t) = \mathfrak{B}(t), \qquad \mathfrak{B}(t) \in \mathbb{R},$$

and the standard Hadamard identity gives

$$C := \mathfrak{B}(0) = 2 + \gamma - \log(4\pi) = 0.046191417932242\ldots. \tag{8.5}$$

For $x > 1$, put

$$S_\Lambda(x) = \sum_{n \leq x} \frac{\Lambda(n)}{n} = \sum_{p^m \leq x} \frac{\log p}{p^m},$$

and, for $0 < z < 1$,

$$T(z) = z \log(1 - z^2) - 2z + \log \frac{1+z}{1-z}.$$

Proposition 14 (Exact finite prime-side formula)**.** *For every $x > 1$,*

$$\boxed{\mathfrak{B}(\log x) = \sqrt{x} \left[S_\Lambda(x) - \log x + \gamma - \left(\frac{\psi(x)}{x} - 1 \right) - \frac{\log(2\pi)}{x} - \frac{1}{2} T(1/x) \right].} \tag{8.6}$$

The expression is continuous at every prime power.

Proof. The standard explicit formulas, initially away from prime powers, are

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2})$$

and

$$S_{\Lambda}(x) = \log x - \gamma + \sum_{\rho} \frac{x^{\rho-1}}{1-\rho} + \sum_{k\geq 1} \frac{x^{-(2k+1)}}{2k+1}.$$

Combining them uses

$$\frac{1}{\rho} + \frac{1}{1-\rho} = \frac{1}{\rho(1-\rho)}.$$

The two trivial-zero contributions combine to $T(1/x)/2$, which proves (8.6). At $x = p^m$, both S_{Λ} and $\psi(x)/x$ jump by $\Lambda(x)/x$; their difference is continuous. □

This formula uses only finite prime-power sums at a given x. It neither generates zeros nor requires CA profiles.

The function $\mathfrak{B}$ is not the Robin margin and is not a replacement for the finite ledger. Its role is precise: equation (5.10) identifies it as the leading normalized spectral mode left by the integrated residual. Controlling that mode subexponentially is already an RH-strength task, as the next theorem makes explicit.

Theorem 15 (Equivalent residual criteria for RH)**.** *The following statements are equivalent:*

1. *the Riemann hypothesis holds;*
2. $\mathfrak{B}$ *is bounded on* $\mathbb{R}$*;*
3. $\mathfrak{B}$ *is positive definite on* $\mathbb{R}$*;*
4. $C - \mathfrak{B}(t) \geq 0$ *for all sufficiently large positive* t*.*

Proof. Under RH, write $\rho = 1/2 + i\gamma_{\rho}$. Then

$$\mathfrak{B}(t) = \sum_{\gamma_{\rho}} \frac{e^{i\gamma_{\rho}t}}{\gamma_{\rho}^2 + 1/4}.$$

For arbitrary complex z_j and real t_j,

$$\sum_{j,k} z_j \overline{z_k} \mathfrak{B}(t_j - t_k) = \sum_{\gamma_{\rho}} \frac{\left|\sum_j z_j e^{i\gamma_{\rho}t_j}\right|^2}{\gamma_{\rho}^2 + 1/4} \geq 0.$$

Hence $\mathfrak{B}$ is positive definite. A 2 by 2 principal minor then gives $|\mathfrak{B}(t)| \leq C$, so positive definiteness implies boundedness.

If $\mathfrak{B}$ is bounded, its Laplace transform is analytic in $\Re s > 0$. Initially in $\Re s > 1/2$,

$$\int_0^{\infty} e^{-st}\mathfrak{B}(t)\,\mathrm{d}t = \sum_{\rho} \frac{1}{\rho(1-\rho)[s - (\rho - 1/2)]}. \tag{8.7}$$

The right-hand side is meromorphic and normally convergent away from its poles, since its high summands are $O(|\gamma_{\rho}|^{-3})$. A zero with $\Re\rho > 1/2$ would give a genuine pole in $\Re s > 0$, contradicting the Laplace transform. The functional equation excludes the reflected zeros.

It remains to treat the one-sided condition. Under RH, pairing ordinates gives

$$C - \mathfrak{B}(t) = \sum_{\gamma_{\rho}>0} \frac{2(1 - \cos(\gamma_{\rho}t))}{\gamma_{\rho}^2 + 1/4} \geq 0. \tag{8.8}$$

Conversely, suppose that $C - \mathfrak{B}(t) \geq 0$ for $t \geq T$. This continuous function has exponential order at most $e^{t/2}$. Apply Landau's theorem for Laplace transforms of nonnegative functions [16, Chapter II, Theorem 5b] to

$$G(s) = \int_0^\infty e^{-su}[C - \mathfrak{B}(T+u)]\,\mathrm{d}u.$$

If $L_C(s)$ denotes the full Laplace transform of $C - \mathfrak{B}$, then

$$G(s) = e^{sT}\left(L_C(s) - \int_0^T e^{-st}[C - \mathfrak{B}(t)]\,\mathrm{d}t\right),$$

so G and L_C have the same positive-half-plane singularities. If its abscissa of convergence were a positive number σ_c, then the real point $s = \sigma_c$ would be singular. On the other hand,

$$\int_0^\infty e^{-st}[C - \mathfrak{B}(t)]\,\mathrm{d}t = \frac{C}{s} - \sum_\rho \frac{1}{\rho(1-\rho)[s - (\rho - 1/2)]}$$

has a meromorphic continuation that is regular at every positive real s: the zeta function has no real zero in $(0,1)$. Hence $\sigma_c \leq 0$, and G is holomorphic in $\Re s > 0$. The identity theorem now excludes every pole with positive real part, proving RH. □

Corollary 16 (Polynomial envelopes are sufficient)**.** *Any bound*

$$|\mathfrak{B}(t)| \leq K(1+t)^A \qquad (t \geq 0)$$

with finite K, A, implies RH. More generally, it is enough that for every $\varepsilon > 0$,

$$|\mathfrak{B}(t)| = O_\varepsilon(e^{\varepsilon t}) \qquad (t \to \infty).$$

Proof. Either bound makes the Laplace transform of $\mathfrak{B}$ holomorphic in every half-plane $\Re s > 0$. The pole-exclusion argument in the proof of Theorem 15 then applies verbatim. □

The underlying positivity principle is not new. Weil's criterion expresses RH through positivity of the Weil distribution [11], and Suzuki proved that pointwise nonnegativity of

$$\Psi_S(t) = \sum_{\lambda \in \mathcal{Z}_\xi} \frac{1 - e^{i\lambda t}}{\lambda^2}$$

is equivalent to RH, where ξ is the completed Riemann xi-function and $\mathcal{Z}_\xi$ is the multiset of zeros of $z \mapsto \xi(1/2 - iz)$, with Suzuki's symmetric convention. His shifted theory also contains an eventual-nonnegativity criterion [15]. The function here is the distinct resolvent-weighted analogue

$$C - \mathfrak{B}(t) = \sum_{\gamma_\rho} \frac{1 - e^{i\gamma_\rho t}}{\gamma_\rho^2 + 1/4} \qquad \text{under RH.}$$

It is not Suzuki's shifted function at parameter 1/2. The contribution of this section is the explicit prime-side cell dynamics and the reduction of its positivity to a prime-power kick inequality, not a new general positivity criterion.

8.3 Exact prime-power cell dynamics

Let $q = p^m$ range over the distinct prime powers and put

$$\tau_q = \log q, \qquad J_q = \frac{\Lambda(q)}{\sqrt{q}}.$$

On an open cell between two consecutive τ_q, set

$$a = \gamma + 1 + S_\Lambda(e^t), \qquad d = \log(2\pi) + \psi(e^t), \qquad z = e^{-t};$$

the two prime sums are constant on that cell. Formula (8.6) becomes

$$\mathfrak{B}(t) = e^{t/2}(a - t) - de^{-t/2} - \frac{1}{2}e^{t/2}T(z). \tag{8.9}$$

Since

$$T'(z) = \log(1 - z^2), \qquad T''(z) = -\frac{2z}{1 - z^2},$$

differentiation gives

$$\mathfrak{B}'(t) = e^{t/2}\left[\frac{a - t}{2} - 1 - \frac{T(z)}{4} + \frac{z}{2}\log(1 - z^2)\right] + \frac{d}{2}e^{-t/2}. \tag{8.10}$$

Theorem 17 (Hybrid cell equation)**.** *On every open prime-power cell,*

$$\boxed{\mathfrak{B}''(t) - \frac{1}{4}\mathfrak{B}(t) = -g(t), \qquad g(t) = e^{t/2}\left(1 - \frac{e^{-3t}}{1 - e^{-2t}}\right).} \tag{8.11}$$

At $q = p^m$,

$$\boxed{\Delta\mathfrak{B}(\tau_q) = 0, \qquad \Delta\mathfrak{B}'(\tau_q) = J_q.} \tag{8.12}$$

Equivalently, in the distributional sense on $(0, \infty)$*,*

$$\mathfrak{B}'' - \frac{1}{4}\mathfrak{B} = -g(t) + \sum_{q=p^m} J_q\delta(t - \tau_q). \tag{8.13}$$

Moreover $g(t) > 0$ and $g'(t) > 0$ for $t \geq \log 2$.

Proof. Equation (8.11) follows by differentiating (8.9). At q, the constants jump by

$$\Delta a = \frac{\Lambda(q)}{q}, \qquad \Delta d = \Lambda(q).$$

Their contributions cancel in $\mathfrak{B}$ and add in $\mathfrak{B}'$, giving (8.12). The signs of g and g' follow directly from $e^{-t} \leq 1/2$. $\square$

There is an important initial-value caveat. Although $\mathfrak{B}(0) = C$,

$$\mathfrak{B}(t) = C + \frac{t}{2}\log t + \frac{\gamma + \log(2\pi) - 1}{2}t + o(t), \qquad t \downarrow 0, \tag{8.14}$$

so $\mathfrak{B}'(0+) = -\infty$. The state must not be initialized as $(C, 0)$. Equivalently, the one-sided quantity in (8.26) has the sharper origin expansion

$$\boxed{\mathfrak{P}(t) = \frac{t}{2}\left[\log\frac{1}{t} + 1 - \gamma - \log(2\pi)\right] + O(t^2)} \qquad (t \downarrow 0). \tag{8.15}$$

8.4 Regularized modes, exact event map, and cell extrema

For $t > 0$, define

$$\widehat{\mathfrak{B}}(t) = \mathfrak{B}(t) + \frac{1}{2}e^{t/2}T(e^{-t}). \tag{8.16}$$

Then

$$\boxed{\widehat{\mathfrak{B}}'' - \frac{1}{4}\widehat{\mathfrak{B}} = -e^{t/2} + \sum_q J_q\delta(t-\tau_q).} \tag{8.17}$$

The removed term is explicit and positive in $\mathfrak{B} - \widehat{\mathfrak{B}}$:

$$\mathfrak{B}(t) - \widehat{\mathfrak{B}}(t) = \sum_{k\geq 1} \frac{e^{-(2k+1/2)t}}{2k(2k+1)} = \frac{1}{6}e^{-5t/2} + O(e^{-9t/2}). \tag{8.18}$$

Thus $\mathfrak{B}$ and $\widehat{\mathfrak{B}}$ have the same polynomial-growth problem. Extending $\widehat{\mathfrak{B}}$ and its first derivative to $t = 0$ by their right limits, the regularized initial data are finite:

$$\widehat{\mathfrak{B}}(0) = \gamma + 1 - \log(2\pi), \qquad \widehat{\mathfrak{B}}'(0) = \frac{\gamma + \log(2\pi) - 1}{2}.$$

The characteristic coordinates are exactly the two classical prime errors:

$$U(t) = e^{-t/2}\left(\widehat{\mathfrak{B}}' + \frac{1}{2}\widehat{\mathfrak{B}}\right) = S_\Lambda(e^t) - t + \gamma, \tag{8.19}$$

$$V(t) = e^{t/2}\left(\widehat{\mathfrak{B}}' - \frac{1}{2}\widehat{\mathfrak{B}}\right) = \psi(e^t) - e^t + \log(2\pi). \tag{8.20}$$

Hence

$$\widehat{\mathfrak{B}} = e^{t/2}U - e^{-t/2}V, \qquad \widehat{\mathfrak{B}}' = \frac{1}{2}\left(e^{t/2}U + e^{-t/2}V\right). \tag{8.21}$$

Between events $U' = -1$, $V' = -e^t$, while at q,

$$\Delta U = \frac{\Lambda(q)}{q}, \qquad \Delta V = \Lambda(q).$$

For consecutive prime powers $q_k < q_{k+1}$, the right states obey the exact two-dimensional recurrence

$$U_{k+1} = U_k - \log\frac{q_{k+1}}{q_k} + \frac{\Lambda(q_{k+1})}{q_{k+1}}, \tag{8.22}$$

$$V_{k+1} = V_k - (q_{k+1} - q_k) + \Lambda(q_{k+1}). \tag{8.23}$$

Partial summation also gives the nonlocal signed identity

$$\boxed{\widehat{\mathfrak{B}}(\log x) = -\sqrt{x}\int_x^\infty \frac{\psi(u) - u}{u^2}\,\mathrm{d}u - \frac{\log(2\pi)}{\sqrt{x}}.} \tag{8.24}$$

It shows explicitly that stability of the residual is a future signed cancellation. Bounding the integrand absolutely loses the required scale.

The cell extrema can also be found without a grid. On a cell with constants a, d, $\widehat{\mathfrak{B}}' = 0$ if and only if

$$e^t(t + 2 - a) = d.$$

Thus there is at most one stationary point,

$$t_* = a - 2 + W_0(de^{2-a}). \tag{8.25}$$

If it lies in the cell and $w = W_0(de^{2-a})$, then

$$\widehat{\mathfrak{B}}(t_*) = 2e^{t_*/2}(1-w), \qquad \widehat{\mathfrak{B}}''(t_*) = -\frac{1}{2}e^{t_*/2}(1+w) < 0.$$

Hence every interior stationary point of $\widehat{\mathfrak{B}}$ is a strict maximum.

The same uniqueness holds for the full $\mathfrak{B}$ when $t \geq \log 2$. Indeed, set

$$h(z) = z + \frac{z}{2}\log(1-z^2) - \frac{1}{2}\log\frac{1+z}{1-z}, \qquad F_B(t) = t + 2 - a - de^{-t} - h(e^{-t}).$$

Then $\mathfrak{B}'(t) = -e^{t/2}F_B(t)/2$, and for $z = e^{-t} \leq 1/2$,

$$0 \leq \frac{d}{dt}h(e^{-t}) = z\left(\frac{2z^2}{1-z^2} - \frac{1}{2}\log(1-z^2)\right) \leq \frac{5}{12}.$$

Consequently

$$F_B'(t) = 1 + de^{-t} - \frac{d}{dt}h(e^{-t}) \geq 1 - \frac{5}{12} > 0.$$

Therefore the full residual also has at most one interior cell stationary point, always a strict maximum. Since every event jumps the derivative upward, events can create minima but never maxima.

8.5 A one-sided first-crossing cone

Put

$$\mathfrak{P}(t) = C - \mathfrak{B}(t), \qquad k(t) = g(t) - \frac{C}{4}. \tag{8.26}$$

For $t \geq \log 2$, $k, k' > 0$, and the hybrid dynamics is

$$\mathfrak{P}'' - \frac{1}{4}\mathfrak{P} = k(t), \qquad \Delta\mathfrak{P}' = -J_q. \tag{8.27}$$

Define, while $\mathfrak{P} > 0$,

$$Q(t) = \sqrt{\frac{\mathfrak{P}(t)^2}{4} + 2k(t)\mathfrak{P}(t)}, \qquad \mathcal{E}_B(t) = \mathfrak{P}'(t)^2 - Q(t)^2. \tag{8.28}$$

On every open cell,

$$\boxed{\mathcal{E}_B'(t) = -2k'(t)\mathfrak{P}(t).} \tag{8.29}$$

At an event,

$$\Delta\mathcal{E}_B = J_q^2 - 2J_q\mathfrak{P}'(\tau_q^-). \tag{8.30}$$

The event increment has no fixed sign. The invariant candidate is the cone $\mathcal{E}_B < 0$, equivalently $|\mathfrak{P}'| < Q$.

Lemma 18 (Initial cone at the first event)**.** *At the right state of the event $q = 2$,*

$$\mathfrak{P}(\log 2) > 0, \qquad \mathcal{E}_B(\log 2+) < 0.$$

Proof. The identity

$$T(1/2) = \frac{3}{2}\log 3 - \log 2 - 1$$

and the prime-side formula give

$$\begin{aligned}
\mathfrak{B}(\log 2) &= \sqrt{2}\left(\gamma + \frac{3}{2} - \log 2 - \frac{1}{2}\log\pi - \frac{3}{4}\log 3\right),\\
\mathfrak{B}'(\log 2-) &= \sqrt{2}\left(\frac{\gamma}{2} - \frac{1}{4} - \frac{1}{2}\log 2 - \frac{1}{8}\log 3 + \frac{1}{4}\log\pi\right),\\
\mathfrak{B}'(\log 2+) &= \mathfrak{B}'(\log 2-) + \frac{\log 2}{\sqrt{2}}.
\end{aligned}$$

Together with $g(\log 2) = 5\sqrt{2}/6$, direct interval evaluation using, for example,

$$0.57721 < \gamma < 0.57722, \qquad 0.69314 < \log 2 < 0.69315,$$
$$1.09861 < \log 3 < 1.09862, \qquad 1.14472 < \log\pi < 1.14474, \qquad 1.41421 < \sqrt{2} < 1.41422$$

gives

$$0.0635 < \mathfrak{P}(\log 2) < 0.0636, \qquad -0.080 < \mathcal{E}_B(\log 2+) < -0.078.$$

The elementary rational bounds above follow from standard series bounds and squaring checks. Thus the initial cone is independent of the exploratory event scan below. □

Theorem 19 (Prime-power kick criterion)**.** *Starting with the right state at $q = 2$, assume the following inductive event condition: whenever the right state at one prime power lies in $\mathfrak{P} > 0$, $\mathcal{E}_B < 0$, then at the next prime power q the pre-event state satisfies*

$$\boxed{Q(\log q) + \mathfrak{P}'(\log q^-) > \frac{\Lambda(q)}{\sqrt{q}}.} \tag{8.31}$$

Then $\mathfrak{P}(t) > 0$ for every $t \geq \log 2$, and RH follows.

Proof. Suppose that $\mathfrak{P} > 0$ and $\mathcal{E}_B < 0$ immediately after an event. Before a hypothetical first zero in the next open cell, (8.29) makes $\mathcal{E}_B$ decrease. At that first zero one would have $\mathcal{E}_B = \mathfrak{P}'^2 \geq 0$, a contradiction. Thus $\mathfrak{P}$ stays positive up to the next event.

Lemma 18 starts the induction. Immediately before the next event, $-Q < \mathfrak{P}' < Q$. The kick changes the derivative to $\mathfrak{P}'_+ = \mathfrak{P}'_- - J_q$. The upper cone boundary is automatic, while the lower boundary is preserved exactly by (8.31). Induction gives eventual positivity of $\mathfrak{P}$; Theorem 15 then gives RH. □

The scalar reserve makes the mechanism especially transparent. Set

$$M_B = Q + \mathfrak{P}', \qquad N_B = Q - \mathfrak{P}', \qquad -\mathcal{E}_B = M_B N_B.$$

Inside the cone $\mathfrak{P} > 0$, $\mathcal{E}_B < 0$, put $A_B = k + \mathfrak{P}/4$. Then

$$\boxed{M_B' = \frac{A_B}{Q} M_B + \frac{k'\mathfrak{P}}{Q} > 0, \qquad M_B^+ = M_B^- - J_q.} \tag{8.32}$$

The smooth flow replenishes the reserve; each prime power withdraws exactly J_q. The missing universal statement is $M_B(\tau_q^-) > J_q$.

There is a fully arithmetic form of the kick. Let

$$\ell(x) = \frac{1}{2}\log(1 - x^{-2}).$$

At a prime power x, the right derivative satisfies

$$\mathfrak{P}'(\log x^+) = \frac{1}{2}(\mathfrak{P} - C) + \frac{x - \psi(x) - \log(2\pi) - \ell(x)}{\sqrt{x}}.$$

Thus the post-kick cone condition is exactly

$$\psi(x) - x < \sqrt{x}\left[Q + \frac{\mathfrak{P} - C}{2}\right] - \log(2\pi) - \ell(x). \tag{8.33}$$

If $\mathfrak{P}$ remains on a fixed positive scale, the leading right-hand side is of order $x^{3/4}$. Present unconditional pointwise PNT envelopes do not yield this bound. A proof must exploit the adaptive state correlation or a block balance; substituting a crude absolute envelope returns to the original obstruction.

8.6 Signed triangular transfer at a prime-power event

The event condition can be rewritten on a short symmetric window without discarding the signs of either the smooth forcing or the nearby prime-power impulses. In distributional notation, (8.27) is

$$D^2\mathfrak{P} = \mathcal{F}(t)\,\mathrm{d}t - \sum_{r=p^a} J_r \delta_{\tau_r}, \qquad \mathcal{F}(t) = \frac{\mathfrak{P}(t)}{4} + k(t). \tag{8.34}$$

Fix a central event $t = \tau_q$, where $q = p^\nu$, and write

$$p^- = \mathfrak{P}'(t^-), \qquad p^+ = \mathfrak{P}'(t^+) = p^- - J_q, \qquad \bar{p} = \frac{p^- + p^+}{2}.$$

For $0 < h < t$, define the signed odd-triangular discrepancy

$$\begin{aligned}
\mathcal{T}^{\triangle}(t,h) = &\sum_{\substack{r=p^a \\ 0<\tau_r - t<h}} (h - (\tau_r - t))\,J_r \\
&- \sum_{\substack{r=p^a \\ 0<t-\tau_r<h}} (h - (t - \tau_r))\,J_r \\
&- \int_0^h (h-s)\big[\mathcal{F}(t+s) - \mathcal{F}(t-s)\big]\,\mathrm{d}s.
\end{aligned} \tag{8.35}$$

The center event is excluded. Endpoint events may be included or excluded, because their triangular weight is zero.

Proposition 20 (Exact signed triangular identity)**.** *At every prime-power event and for every* $0 < h < t$*,*

$$\boxed{2h\bar{p} = \mathfrak{P}(t+h) - \mathfrak{P}(t-h) + \mathcal{T}^{\triangle}(t,h).} \tag{8.36}$$

Consequently,

$$\boxed{2hp^+ = \mathfrak{P}(t+h) - \mathfrak{P}(t-h) + \mathcal{T}^{\triangle}(t,h) - hJ_q.} \tag{8.37}$$

Proof. Integrate the first derivative from the center separately to the right and to the left, and then substitute the measure equation (8.34). A right-hand impulse lowers the right endpoint by its remaining triangular weight and therefore appears with a positive sign in $\mathcal{T}^{\triangle}$; a left-hand impulse has the opposite sign. The central atom has zero odd-triangular weight and is already encoded by the midpoint slope. Finally use $\bar{p} = p^+ + J_q/2$. ☐

Proposition 21 (Exact short-window lower-cone transfer)**.** *Suppose that the pre-event state is in the cone* $\mathfrak{P}(t) > 0$, $|p^-| < Q(t)$*. Then the post-event state remains in that cone if and only if*

$$\boxed{\mathfrak{P}(t+h) - \mathfrak{P}(t-h) + \mathcal{T}^{\triangle}(t,h) + 2hQ(t) - hJ_q > 0.} \tag{8.38}$$

Proof. The downward kick makes $p^+ < p^- < Q(t)$, so the upper cone boundary is automatic. The lower boundary is $p^+ > -Q(t)$, which is equivalent to (8.38) by (8.37). ☐

Thus (8.38) is not another approximation to the kick criterion: under the inherited cone it is exactly the same event condition, expressed entirely through a signed prime-side window. A natural adaptive choice is

$$h = h_\lambda(t) = \lambda t e^{-t/2} = \lambda \frac{\log q}{\sqrt{q}}. \tag{8.39}$$

For $q = p^\nu$,

$$\frac{J_q}{h} = \frac{1}{\lambda\nu}, \qquad hJ_q = \frac{\lambda(\log q)^2}{\nu q}. \tag{8.40}$$

The multiplicative window is $qe^{-h} < r < qe^h$, with additive half-width $qh \sim \lambda\sqrt{q}\log q$.

8.7 RH-conditional Turán calibration and the critical window

The symmetric secant in (8.37) has a natural spectral bound, but its logical status is essential.

Proposition 22 (RH-conditional central-secant bound)**.** *Assume RH. Then for every real t, h,*

$$\boxed{|\mathfrak{P}(t+h) - \mathfrak{P}(t-h)|^2 \le 4\mathfrak{P}(t)\mathfrak{P}(2h).} \tag{8.41}$$

Proof. Under RH, (8.8) becomes

$$\mathfrak{P}(t) = 2\sum_{\gamma>0} \frac{1-\cos(\gamma t)}{\gamma^2 + 1/4}.$$

The centered difference is

$$4\sum_{\gamma>0} \frac{\sin(\gamma t)\sin(\gamma h)}{\gamma^2 + 1/4}.$$

Weighted Cauchy–Schwarz, together with

$$\sum_{\gamma>0} \frac{\sin^2(\gamma h)}{\gamma^2 + 1/4} = \frac{\mathfrak{P}(2h)}{4}, \qquad \sum_{\gamma>0} \frac{\sin^2(\gamma t)}{\gamma^2 + 1/4} \le \mathfrak{P}(t),$$

proves the claim. □

Put

$$D_h = \frac{\mathfrak{P}(t+h) - \mathfrak{P}(t-h)}{2h}, \qquad E_h = \bar{p} - D_h = \frac{\mathcal{T}^{\triangle}(t,h)}{2h}, \tag{8.42}$$

$$S_h = \frac{\sqrt{\mathfrak{P}(t)\mathfrak{P}(2h)}}{h}, \qquad U_\lambda(q) = 4\mathfrak{P}(t)\mathfrak{P}(2h) - \big[\mathfrak{P}(t+h) - \mathfrak{P}(t-h)\big]^2, \tag{8.43}$$

$$M_\lambda(q) = Q(t) - S_h + E_h - \frac{J_q}{2}. \tag{8.44}$$

At any event where $\mathfrak{P}(t), \mathfrak{P}(2h) > 0$, the two inequalities

$$U_\lambda(q) \ge 0, \qquad M_\lambda(q) > 0 \tag{8.45}$$

imply $Q(t) + p^+ > 0$: indeed $D_h \ge -S_h$ and $p^+ = D_h + E_h - J_q/2$. Under RH the first inequality follows from Proposition 22. In an unconditional proof, however, (8.45) is an open two-lemma target, not an available estimate.

The scale (8.39) explains the numerical constant $1/4$. From (8.15), for large t,

$$\frac{\mathfrak{P}(2h)}{h^2} = \frac{e^{t/2}}{\lambda t}\left[\frac{t}{2} - \log(2\lambda t) + 1 - \gamma - \log(2\pi) + O(h)\right]. \tag{8.46}$$

On the other hand,

$$\frac{Q(t)^2}{\mathfrak{P}(t)} = 2e^{t/2} - \frac{C}{2} + \frac{\mathfrak{P}(t)}{4} + O(e^{-5t/2}).$$

Thus $\lambda > 1/4$ leaves a fixed leading margin between the cone radius and the RH-conditional worst secant, while $\lambda < 1/4$ does not. At the critical value $\lambda = 1/4$, in the RH-conditional bounded positive $\mathfrak{P}$ regime,

$$\boxed{\frac{Q(t)^2}{\mathfrak{P}(t)} - \frac{\mathfrak{P}(2h)}{h^2} = \frac{4e^{t/2}}{t}\big[\log(\pi t) + \gamma - 1\big] - \frac{C}{2} + \frac{\mathfrak{P}(t)}{4} + O(1) > 0} \tag{8.47}$$

for all sufficiently large t. This positive next-order correction is a calibration only: it still uses the RH-conditional secant inequality and says nothing by itself about the sign of E_h.

There is a sharp logical obstruction to importing (8.41) as a routine short-interval estimate. Since $\mathfrak{P}(2h) > 0$ for sufficiently small h, eventual validity of the adaptive family $U_\lambda(q) \geq 0$, or of its continuous analogue for all large t, forces eventual nonnegativity of $\mathfrak{P}(t)$ and hence RH by Theorem 15. Conversely RH proves the family. The distributional equation alone cannot supply this positivity: adding $A\sinh(t/2)$ preserves its forcing and impulses while allowing the Turán inequality to fail for large t.

Nor is decay of the triangular discrepancy alone sufficient. For a generic exponential mode $f_\mu(t) = e^{\mu t}$, its exact triangular remainder is

$$\begin{aligned}\mathcal{T}_\mu(t,h) &= 2hf_\mu'(t) - \big[f_\mu(t+h) - f_\mu(t-h)\big] \\ &= 2e^{\mu t}\big[\mu h - \sinh(\mu h)\big] \\ &= -\frac{\mu^3h^3}{3}e^{\mu t} + O(h^5e^{\mu t}).\end{aligned} \tag{8.48}$$

Therefore

$$\frac{\mathcal{T}_\mu(t,h)}{2hf_\mu'(t)} = O(h^2) \longrightarrow 0 \tag{8.49}$$

even when $\Re\mu > 0$. A growing off-critical mode can have a rapidly weakening relative remainder. Any successful argument must control the central secant together with the signed impulse discrepancy, or prove the combined inequality (8.38) directly.

The target window has additive scale $\sqrt{x}\log x = x^{1/2+o(1)}$. Current unconditional uniform PNT technology reaches the longer scale $x^{17/30+o(1)}$ [13]; RH-conditional explicit square-root-scale results are absolute short-interval estimates [14], not the odd reserve-adaptive inequality above. Ford–Soundararajan–Zaharescu connect zero statistics, pair correlation, and primes in short intervals [12]. Their framework is relevant to an alternative zero-dephasing route, but it is not the signed triangular event identity or a proof of (8.38).

8.8 Polynomial energy audit

The regularized state suggests a time-dependent nearly degenerate energy. For $r \geq 0$, put

$$\varepsilon_r(t) = (1+t)^{-r}$$

and

$$E_r(t) = \left(1 - \frac{\varepsilon_r(t)}{2}\right)\widehat{\mathfrak{B}}(t)^2 + 2\varepsilon_r(t)\widehat{\mathfrak{B}}'(t)^2. \tag{8.50}$$

On an open cell, exact differentiation gives

$$E_r' = 2\widehat{\mathfrak{B}}\widehat{\mathfrak{B}}' + \varepsilon_r'\left(2\widehat{\mathfrak{B}}'^2 - \frac{1}{2}\widehat{\mathfrak{B}}^2\right) - 4\varepsilon_r e^{t/2}\widehat{\mathfrak{B}}'. \tag{8.51}$$

At a prime-power event,

$$\boxed{\Delta E_r = 4\varepsilon_r J_q\widehat{\mathfrak{B}}'_- + 2\varepsilon_r J_q^2.} \tag{8.52}$$

Proposition 23 (Square-kick budget)**.** *The positive quadratic kick budget satisfies*

$$\boxed{\sum_{q=p^m}(1+\log q)^{-r}\frac{\Lambda(q)^2}{q}<\infty \quad\iff\quad r>2.} \tag{8.53}$$

Proof. The layers $m \geq 2$ converge for every fixed r, since the geometric factor p^{-m} dominates the logarithmic weight. The prime layer is

$$\sum_p \frac{(\log p)^2}{p(1+\log p)^r}.$$

The PNT and partial summation give the same convergence behavior as

$$\int^\infty \frac{(\log u)^{1-r}}{u}\,\mathrm{d}u = \int^\infty v^{1-r}\,\mathrm{d}v,$$

which converges exactly for $r > 2$. □

This makes the unavoidable positive square cost finite, but the signed work $4\varepsilon_r J_q \widehat{\mathfrak{B}}'_-$ remains. Equation (8.52) is therefore a concrete block-cancellation target, not a proved polynomial envelope.

There is also a structural no-go for a naive universal Lyapunov argument. The homogeneous cell flow has an unstable mode $e^{t/2}$, so no fixed positive-definite quadratic form decreases for every state of that flow. In the normalized characteristic variables

$$\alpha = \widehat{\mathfrak{B}}' + \frac{1}{2}\widehat{\mathfrak{B}}, \qquad \beta = \widehat{\mathfrak{B}}' - \frac{1}{2}\widehat{\mathfrak{B}},$$

every event is the common translation $(\alpha, \beta) \mapsto (\alpha + J_q, \beta + J_q)$. A positive semidefinite quadratic form invariant under all such translations must annihilate $(1, 1)$, and hence degenerates to a multiple of $(\alpha - \beta)^2 = \widehat{\mathfrak{B}}^2$. This does not rule out an energy adapted to the actual arithmetic trajectory; it explains why the prime-state correlation cannot be discarded.

8.9 Exploratory finite diagnostics

An event-driven binary64 implementation scanned every prime-power event in

$$2 \leq q \leq 56\,048\,351.$$

It processed 3 342 115 events and located 23 746 interior stationary maxima of the full $\mathfrak{B}$. The observed extrema were

$$\min \mathfrak{B} = -0.0226812404648 \quad (q = 319\,439)$$

and

$$\max \mathfrak{B} = 0.0246942372460$$

at the interior point $x \approx 34\,186\,454.4254$. Thus the largest observed absolute value was below C, but only on this finite range.

Both one-sided event states remained inside $\mathcal{E}_B < 0$. The smallest pre-kick ratio was

$$\min_q \frac{Q(\tau_q) + \mathfrak{P}'(\tau_q^-)}{J_q} = 1.205548185874075\ldots \quad (q = 5), \tag{8.54}$$

and the smallest post-kick reserve was

$$0.121238569235683\ldots \quad (q = 2).$$

About half the individual energy kicks were positive. For every tested $r \in \{1, 2, 3, 4, 6, 8\}$, the total signed change of E_r over the scanned event range was negative, while local monotonicity failed. Interior extrema of E_r were not searched.

The maximum validation residuals were approximately

$$1.46 \times 10^{-11} \quad \text{for continuity,} \qquad 7.50 \times 10^{-12} \quad \text{for the derivative jump,}$$

and 1.75×10^{-11} against an independent evaluation of (8.6). These are consistency diagnostics in ordinary floating-point arithmetic, not outward-rounded enclosures. They support the choice of invariant but prove nothing about the infinite sequence of kicks.

8.10 Exploratory signed-triangular diagnostics

A second binary64 experiment evaluated the quantities (8.42)–(8.44) at every distinct prime-power center

$$q = p^a \leq 10^8$$

for $\lambda \in \{0.25, 0.30, 0.50, 1.00\}$. The population consisted of 5 761 455 prime events and 1 404 higher prime-power events, hence 5 762 859 centers. The endpoint evaluator used prime-power support through 100 184 379. The archived run used CPython 3.13.5, NumPy 2.3.5, Windows 11, and IEEE binary64 arithmetic.

No negative binary64 value of $U_\lambda(q)$ occurred among the $4 \times 5\,762\,859$ event-window evaluations. The principal finite statistics were

λ	$\min U_\lambda$	event q	$\min U_\lambda/(4\mathfrak{P}(t)\mathfrak{P}(2h))$	$\#\{M_\lambda < 0\}$	last q
0.25	3.2533463×10^{-4}	34 186 693	0.9044830274	7	19
0.30	3.7648594×10^{-4}	34 186 693	0.9047124058	5	19
0.50	5.6256791×10^{-4}	34 186 693	0.8835097611	3	5
1.00	9.4955723×10^{-4}	34 186 619	0.8960333801	0	–

For $\lambda = 1/4$, $M_\lambda(q) > 0$ at every scanned center $23 \leq q \leq 10^8$; its minimum in $10^7 \leq q < 10^8$ was 3.6226194326. For $\lambda = 1$, it was positive throughout the scan, with finite minimum 0.0251158153123 at $q = 2$. These are statements about the stored finite scan only.

The sign of E_h was nearly balanced. At $\lambda = 1/4$, the observed decade maxima of the radius-normalized discrepancy were

q range	$10^2 : 10^3$	$10^3 : 10^4$	$10^4 : 10^5$	$10^5 : 10^6$	$10^6 : 10^7$	$10^7 : 10^8$
$\max \lvert E_h \rvert / Q$	0.3141952	0.2672301	0.1053174	0.0505864	0.0213857	0.0078766

A descriptive log–log fit to the last four maxima had slope -0.369607 and $R^2 = 0.985$. This is a finite-range empirical pattern, not a polynomial envelope, a monotonicity theorem, or evidence that $E_h = o(Q)$ has been proved. In particular, (8.49) shows why such weakening alone could not exclude an off-line exponential mode.

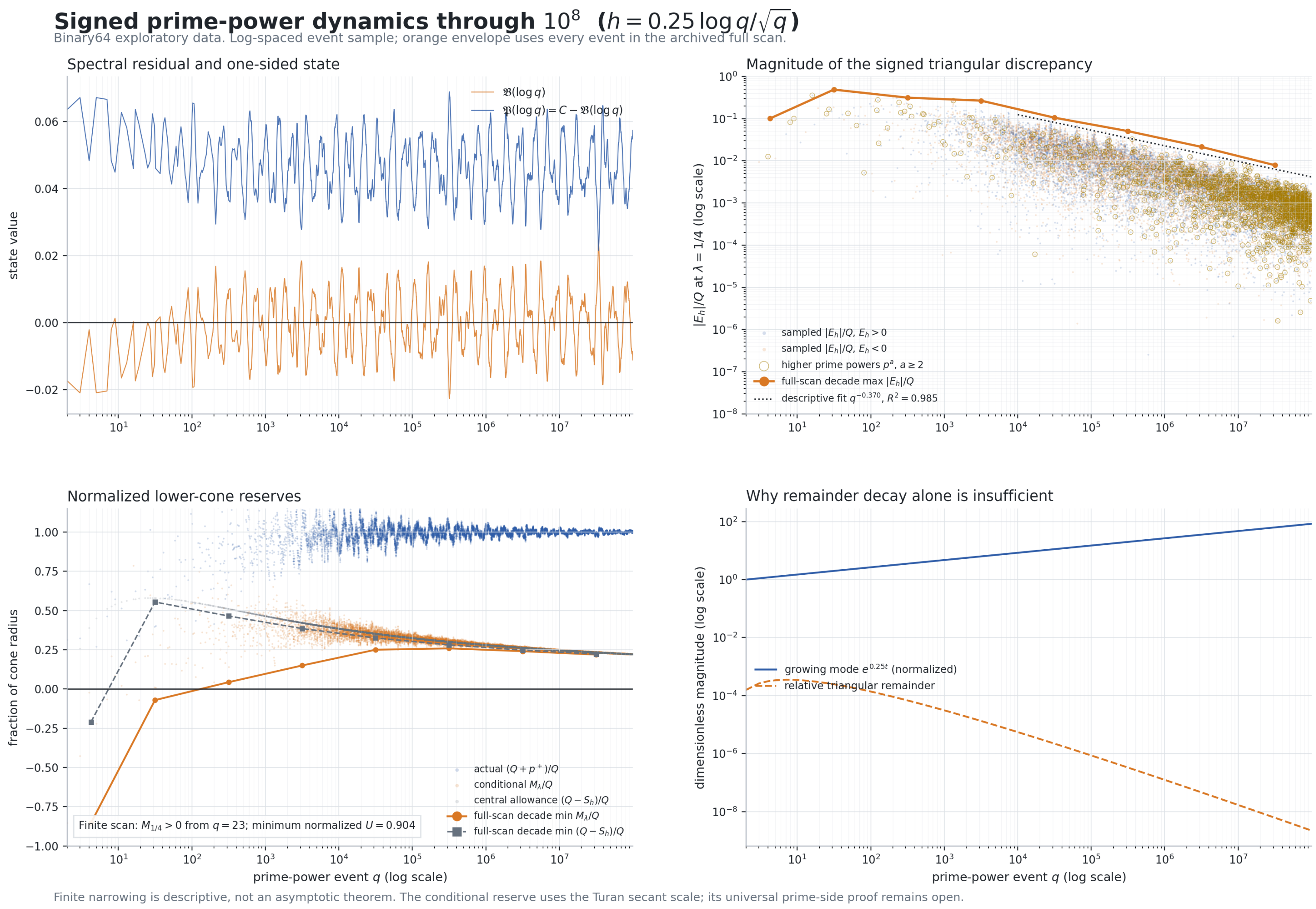

Figure 1: Signed prime-power dynamics at $\lambda = 1/4$ through $q \leq 10^8$. The upper panels show sampled states and the signed triangular discrepancy, with orange maxima from the full scan. The lower panels compare normalized reserves and an analytic off-line test mode. The arithmetic panels are exploratory binary64 data, not interval certificates or asymptotic theorems.

Figure 1 separates facts that must not be conflated. The upper-left panel displays a deterministic sample of 17 836 prime-power events: logarithmically spaced centers, every early event through 10^4, and every higher prime power through 10^8. The curves are joined only to make the oscillatory shape visible. The orange envelope in the upper-right panel is stronger than the plotted sample: each point is the maximum of $|E_h|/Q$ over an entire decimal decade of the full 5 762 859-event scan. Its fitted exponent -0.369607 is descriptive on this finite range only.

The lower-left panel compares the actual quantity needed after an event with the modular RH-conditional lower bound. Its finite positivity from the next event $q = 23$, after the last failure at $q = 19$, is consistent with the cone mechanism, but the interpretation of $M_{1/4}$ as a lower bound still requires $U_{1/4} \geq 0$. The lower-right panel is not arithmetic data. It plots the exact analytic test $f_\mu(t) = e^{\mu t}$ with $\mu = 1/4$. While $f_\mu(t) = q^{1/4}$ grows, (8.49) gives

$$\frac{|\mathcal{T}_\mu(t,h)|}{2h|f'_\mu(t)|} = \frac{\sinh(\mu h)}{\mu h} - 1 \sim \frac{\mu^2\lambda^2(\log q)^2}{6q} \longrightarrow 0.$$

Thus a shrinking relative triangular remainder measures local linearity on a shrinking window; by itself it does not exclude a globally growing off-line mode. The infinite target remains the full combined condition (8.38), or an unconditional proof of both members of (8.45).

An implementation independent of the scanner reconstructed the prime-power support and directly integrated the atomic and regular triangular terms. At $q = 997$, $\lambda = 1/2$, it obtained

$$\mathcal{T}^{\triangle}(t,h) = 2hE_h = 0.012980934864669306848796093728077777752292\ldots$$

with an 80-digit residual of 7.96×10^{-79}. Ten tests passed with `mpmath` 1.3.0: eight scanner regression tests and two independent direct-QA tests. High-precision spot checks retained the signs of the reported extremal values. None of these calculations is an interval certificate, and finite positivity of U_λ does not prove its RH-equivalent universal extension.

9 Executable certificates and reproducibility

9.1 Certificate summary

certificate	result
ordinary-integer base	50 400 integers, no failure
support-11 overlap	exact common endpoint 55 440
small CA bridge	503 profiles, 44 ties, no failure
direct CA scan	3 341 978 profiles, 1 427 ties, no failure
finite D_* scan	3 340 551 supports, no inconclusive case
extended residual cover	662 segments, positive minimum
extended precision audit	4 007 matching intervals nested
all-integer cutoff audit	7 matching intervals nested

All paths below are relative to the root of the public software repository [18]. The complete companion directory is `https://github.com/robopol/Riemann-hypothesis/tree/main/verification/finite_robin_prime_power`. The canonical certifiers and reports are:

purpose	artifact and SHA-256
direct all-profile scan	`verification/finite_robin_prime_power/test_ca_all_profile_interval_certificate.py` `8da8ad3d63b6dec8ffd4838b968a0added4c26eb5810fa2969dadb917f691077`

purpose	artifact and SHA-256
	verification/finite_robin_prime_power/ca_all_profile_ interval_certificate_report.json 8ac1ee276df64cc91d93b483b501d34f8a9b69408d2fb065 0d4bab8f43304f83
finite D_* sweep	verification/finite_robin_prime_power/test_ca_exact_ buffer_interval_certifier.py add8f04a2e02b606491c1bdf6aab4fcb23f5d10e2db20232 6fded0c5035d5ef2 verification/finite_robin_prime_power/ca_exact_buffer_ interval_certificate_report.json 57c14d33aa47b7ddff012043304151fc9b11b50a114c2634 665f7455ce80dba1
analytic D_* continuation	verification/finite_robin_prime_power/test_asymptotic_ dstar_lower_bound.py c7981a2fdec34211e07434d82931c2ac534a20e384fdace3 1b4ee6d79a8211b5 verification/finite_robin_prime_power/asymptotic_dstar_ lower_bound_report.json ea33332d7acade20b35409131a6a5acac00d722cf3439d9b 36cb7992d296fa18
extended residual	verification/finite_robin_prime_power/test_ca_residual_ upward_extended_certificate.py 165f7a069c93ede79709ed559fdd1a9908e41230fd4b5208 44931789a97d3e95 verification/finite_robin_prime_power/ca_residual_upward_ extended_certificate_report.json 938713a1e1443196e349269d9f88f429d3859aa262fbf110 42733c6ef1278263 verification/finite_robin_prime_power/ca_residual_upward_ extended_certificate_p80_report.json 41244347c6366c3321dc653235b9db32008e38678654d2f8 99838b7f5c2ee8a1
integer base	verification/finite_robin_prime_power/test_robin_base_ 5041_55440_interval.py ee296aa4a9e57e576da69a114e5983d9dad172d89e35d149 8c83a1cff211c1aa verification/finite_robin_prime_power/robin_base_5041_ 55440_interval_report.json 161381cb69f55030f2f6bb0c57c6518383aa66a7d65165cf 98e1686aa013cf45
support-11 overlap	verification/finite_robin_prime_power/test_ca_support_11_ overlap_certificate.py 660420eba61fa0ad61df16ed6117dfd76ac84adb78cff289 2c44e0b500de29a9 verification/finite_robin_prime_power/ca_support_11_ overlap_certificate_report.json af903d574499c2b9450b9afed9b4799d3d01a668eb300973 06d61ba97da1a0bd
small CA bridge	verification/finite_robin_prime_power/test_ca_all_profile_ interval_certificate.py 8da8ad3d63b6dec8ffd4838b968a0added4c26eb5810fa29 69dadb917f691077 verification/finite_robin_prime_power/ca_all_profile_ bridge_11_3299_report.json

purpose	artifact and SHA-256
	e29d66dde12af0e773d86827b0237ed323f62abfd19bc7fe 028a7e065782e5ca
all-integer cutoff	verification/finite_robin_prime_power/test_ca_all_integer_cutoff_certificate.py df1238bee0184a3dccebdcaf72920b9362984ec845273692 a786bf69030e4052 verification/finite_robin_prime_power/ca_all_integer_cutoff_certificate_report.json 0ba63f2645fc5c312aa00890daff8b87b720e68656c5cddf c31dd1465284f8f3 verification/finite_robin_prime_power/ca_all_integer_cutoff_certificate_p80_report.json 5dde972048d365efe11a9305bec698a0f0c56de0e27957f8 d72f67a72b00351e
paper manifest and precision audit	verification/finite_robin_prime_power/test_finite_robin_paper_manifest.py 5fb26886161b3aa64ded0c57b49dfd4265a47f59b2b220c0 b0586c141512faa3

9.2 Exploratory signed-dynamics artifacts

The following artifacts support only the binary64 diagnostics and independent numerical QA in Section 8. They are deliberately separated from the directed-interval certificates above.

purpose	artifact and SHA-256
event-driven implementation	verification/finite_robin_prime_power/exploratory_b_cell_dynamics.py 1f1722248765f2ba314fbb5c17cd7727a58ef363f00254ed 9ff9edb0fbd68c21
full finite scan report	verification/finite_robin_prime_power/exploratory_b_cell_dynamics_report.json 5c610b2e08ccc94b30b158d1cd578dafad3189042491aa4e 6d7006670c25da41
moderate-range regression test	verification/finite_robin_prime_power/test_exploratory_b_cell_dynamics.py 446f0e22fe617c1ed7809dd3d2e78f1a10420cd611be4c8b 33239ceb5e17b773
derivation and experiment log	verification/finite_robin_prime_power/B_Prime_Power_Cell_Dynamics_Investigation_2026-08-04.md acb81a8c38101b3e5fada926faf75ac24a05e67d7d231566 7c6aac49bff22b8b
signed-triangular scanner	verification/finite_robin_prime_power/exploratory_signed_triangular_scan.py d03e166a4bd5fde6a9ed8ed583ab74df31db7d0fc214090b d1fb9160519b9daa
eventwise prime-side helper	verification/finite_robin_prime_power/exploratory_eventwise_reserve_scan.py 75f4889ae5133f2571e05233394197fb09545f5d5d0ebc75 163707521bc5e77b
signed-triangular 10^8 report	verification/finite_robin_prime_power/exploratory_signed_triangular_scan_1e8_report.json

purpose	artifact and SHA-256
	b464e154ee46a8caf9a223dda8c60ebcacd17eefa4d9548c 33b10e16232743a4
signed-triangular regression test	verification/finite_robin_prime_power/test_exploratory_signed_triangular_scan.py ab61a564273346f838af4d13498b02fb3f2e01d078f2b32f 030fe86d6d1672f7
independent triangular direct QA	verification/finite_robin_prime_power/test_exploratory_signed_triangular_direct_qa.py b53b528e32d09f27ad1d9f35dbae9800d0f665ce508eaaba cc7d119e96fdae92
signed-dynamics figure generator	verification/finite_robin_prime_power/plot_signed_triangular_dynamics.py 2ede87d1f57dea113ad9d24ee396c52786c3261b5357c3b1 c29e03ff975b9237
paper figure, raster	papers/figures/signed_triangular_dynamics_lambda_0p25.png 3bcbf276c578c61fed17548ac66081414caa9238fff8dcec 78e9c37623cdb48a
paper figure, vector	papers/figures/signed_triangular_dynamics_lambda_0p25.svg 114f1fd10ccc8338d90bbdce8e85e0138ae0ffaf3517a562 91422e19113b092b
plotted deterministic sample	papers/figures/signed_triangular_dynamics_lambda_0p25_sample.csv e35fd4e97c829f5310bdba29d5d668a83f9bcb7b1d7ed40e 9a2c695b9bd218bb

The common prime-stream SHA-256 through 56 048 351 is

`9438df02de33467c1ed4307c0a801c2c57777e00bb7413b17ce955a117c8b2c9`.

The top-level cutoff certifier reopens the canonical dependency reports, checks their domains and PASS fields, and verifies all direct plus six transitive source/report hashes against the current files.

9.3 Integrity audit and recomputation

```
Set-Location verification/finite_robin_prime_power

python test_finite_robin_paper_manifest.py
python test_exploratory_b_cell_dynamics.py
python test_exploratory_signed_triangular_scan.py
python test_exploratory_signed_triangular_direct_qa.py
```

The first command is the canonical non-destructive audit of the archived release: it reopens the reports, verifies their listed hashes and headline values, and checks all (4,007+7) cross-precision interval inclusions. The second command is a moderate-range regression test of the exploratory prime-power implementation. The final two commands test the signed triangular scanner and its independent direct reconstruction; the 80-digit branch of the last command requires `mpmath`.

Exact parameters for fresh recomputation, including the full direct sweep, the binary64 cell scan through 56 048 351, the signed-triangular scan through 10^8, and deterministic reproduction of Figure 1, are recorded in `verification/finite_robin_prime_power/README.md`. Fresh runs must write to alternate report names or be made in a disposable copy. The archived JSON files deliberately record environment-dependent metadata such as absolute paths, command-line arguments, and timings, so a newly generated report is expected to have a different whole-file SHA-256 even when its certified mathematical fields agree. A release intended for external verification should archive the complete repository under an immutable commit and a persistent DOI; branch-relative paths alone are not a permanent citation.

9.4 Directed arithmetic and trusted base

The interval implementations use separate floor and ceiling decimal contexts for arithmetic operations. Logarithms and square roots are evaluated at higher precision and padded outward by explicit ulps. Euler's constant is self-enclosed by harmonic-number inequalities, and π is self-enclosed by Machin's formula where required. Integer divisor sums, prime powers, threshold indices, and sieve state use exact Python integers.

The trusted base consists of:

- the analytic lemmas stated in this paper;
- the cited published theorems;
- CPython integer and Decimal semantics in the recorded environment;
- correctness of the deterministic sieve and certificate programs;
- correctness of the operating system and hardware execution.

Independent 60- and 80-digit runs give nested enclosures. This substantially exceeds ordinary floating-point evidence, but it is not a formal verification of the implementation.

10 Present barrier and rigor boundary

The same-point scalar envelope used in the upper certificate is positive at

$$1.64967 \times 10^{23}$$

by at least $3.2833942493 \times 10^{-7}$, but is negative at

$$1.64968 \times 10^{23}$$

by more than $1.5193588576 \times 10^{-6}$. Nothing singular happens to a CA profile at this decimal endpoint. The sign change belongs only to the absolute lower envelope.

For fixed verified height T, the dominant term in (5.14) grows essentially as

$$S_2(T)\sqrt{x}.$$

In contrast, the explicit lower reserve $D_{\mathrm{lb}}(x)$ approaches a finite constant. Therefore the present absolute high-zero treatment must eventually exhaust the deterministic reserve.

There are now two distinct continuation routes. To extend the *finite certificate*, one may raise the rigorous zero-verification height, sharpen the directed high-zero or off-line-zero estimates, or change the smoothing kernel. To address the *universal quantifier*, the signed formulation in Section 8 gives a more concrete target: prove the eventwise reserve inequality (8.31); equivalently, prove the sharp combined signed-window inequality (8.38). A stronger modular route is the pair (8.45), and a fourth possibility is a block estimate controlling the signed cross-work in (8.52). The Turán member of the modular pair is itself RH-strength and is presently available here only under RH. Thus a tighter absolute envelope is not the only structural route, although it remains the mechanism behind the finite theorem proved here.

The following points delimit the claim.

1. No full-RH assumption is made. The Platt–Trudgian result is a published finite-height theorem.
2. This paper does not use a universal assertion $\theta(p) < p$. The primary layer bound uses the global Rosser–Schoenfeld estimate; an optional stronger layer audit uses Büthe only in its stated finite range. The support-to-size conversion uses the lower Dusart estimate (4.1).
3. The negative scalar envelope above the endpoint is not a counterexample to Robin's inequality.
4. The signed buffer identity, cell equation, cone criterion, and exact triangular transfer are unconditional. The Turán secant bound in Proposition 22 explicitly assumes RH and is not used in either finite theorem.
5. The binary64 scans through 56 048 351 and 10^8 are exploratory. They do not establish a universal kick criterion, a universal Turán defect, or an asymptotic bound for E_h/Q.
6. The result proves nothing for integers above $10^{7.1\times 10^{22}}$, and it does not prove the Riemann hypothesis.

11 Conclusion

The full-support CA ledger admits the exact prime-power reduction

$$\mathcal{G}(n) = \mathcal{R}_\infty(x) + \mathcal{B}_2(n,x) + R_{\mathrm{core}}(n,x) - \mathcal{C}_{\mathrm{pp}}(x),$$

with a strictly positive, profile-independent deterministic buffer $D_*(x)$. An exhaustive CA profile computation closes the lower finite support range, while a directed analytic residual certificate continues the result to 1.64967×10^{23}. Robin's consecutive-CA convexity proposition and an explicit support-to-size estimate then give the all-integer cutoff

$$5041 \le n \le 10^{7.1\times 10^{22}}.$$

The remaining obstruction is not the finite prime-power reduction or the CA handoff. For the present finite certificate it is the absolute treatment of the unverified high-zero tail beyond the endpoint. At the signed level, the same obstruction has been reduced to an exact hybrid system: smooth cells replenish the cone reserve M_B, and each prime power q withdraws exactly $\Lambda(q)/\sqrt{q}$. The initial cone is elementary, and the remaining universal target is the prime-power inequality (8.31). The signed triangular identity rewrites that same lower-cone transfer as the sharp short-window inequality (8.38) on the natural additive scale $\sqrt{q}\log q$. RH conditionally supplies a Turán secant bound whose critical window is $h = (\log q)/(4\sqrt{q})$, including a favorable next-order correction. The scan through 10^8 finds no negative Turán defect and finds a positive conditional reserve after a small finite base, but it is ordinary floating-point evidence only. Moreover, the exponential mode calculation (8.48) proves that decay of the triangular discrepancy alone cannot exclude an off-line mode. What remains is an unconditional prime-side proof of the combined inequality, the two-lemma target, or an adequate block signed-work estimate. None is proved here, so the conclusion remains a finite verification rather than a proof of RH.

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